



RESEARCH ARTICLE

Long-term nutrient management reshapes soil carbon fractions and biological fertility: Evidence from a 43-year semi-arid Vertisol experiment

S Kaviya Sri¹, B Bhakiyathu Saliha^{2*}, Sathiya Bama Kaliappan¹, Boomiraj Kovilpillai³, Sumathi C S⁴, G Guru⁵, Sanjivkumar V², M Manikandan², J V N S Prasad⁶, V K Singh⁷ & Priya Pandurang Gurav⁶

¹Department of Soil Science and Agricultural Chemistry, Agricultural College and Research Institute, Tamil Nadu Agricultural University (TNAU), Coimbatore 641 003, Tamil Nadu, India

²Agricultural Research Station, Tamil Nadu Agricultural University (TNAU), Kovilpatti 628 501, Tamil Nadu, India

³Department of Environmental Sciences, Agricultural College and Research Institute, Tamil Nadu Agricultural University (TNAU), Coimbatore 641 003, Tamil Nadu, India

⁴Department of Physical Sciences and Information Technology, Agricultural College and Research Institute, Tamil Nadu Agricultural University (TNAU), Coimbatore 641 003, Tamil Nadu, India

⁵Department of Agronomy, Agricultural College and Research Institute, Tamil Nadu Agricultural University (TNAU), Coimbatore 641 003, Tamil Nadu, India

⁶All India Coordinated Research Project for Dryland Agriculture, ICAR-Central Research Institute for Dryland Agriculture, Hyderabad 500 059, Telangana, India

⁷ICAR-Central Research Institute for Dryland Agriculture (CRIDA), Santoshnagar, Hyderabad 500 059, Telangana, India

*Correspondence email - bhakiyathusaliha@tnau.ac.in

Received: 30 April 2026; Accepted: 10 July 2026; Available online: Version 1.0: 07 September 2026

Cite this article: Kaviya SS, Bhakiyathu SB, Sathiya BK, Boomiraj K, Sumathi CS, Guru G, Sanjivkumar V, Manikandan M, Prasad JVNS, Singh VK, Priya PG. Long-term nutrient management reshapes soil carbon fractions and biological fertility: Evidence from a 43-year semi-arid Vertisol experiment. *Plant Science Today*. 2026;13(sp5):01–17. <https://doi.org/10.14719/pst.15338>

Abstract

Improving soil carbon quality and biological fertility through long-term nutrient management is essential for sustaining productivity, reducing input dependence and enhancing climate resilience in rainfed Vertisols. This study evaluated the effects of nine treatments including control, inorganic fertilisation (50 % RDF, 100 % RDF, RDF + Zn), organic amendments [farmyard manure (FYM) (50 % N_{FYM}, sole FYM), crop residues (CR) (50 % N_{CR})] and integrated nutrient management (INM_{FYM}, INM_{CR}) on soil carbon pools, oxidisable carbon fractions, carbon management indices (CMI) and enzyme activities in a permanent manurial experiment. Soil organic carbon (SOC) ranged from 5.66 (control) to 11.19 g kg⁻¹ (INM_{FYM}), both in the cotton phase. Inorganic treatments enhanced soil inorganic carbon (SIC), suggesting an inverse relationship with organic inputs. Sole FYM enriched the very labile fraction, whereas INM enhanced the labile and less labile fractions and non-labile carbon was highest under RDF + Zn. CMI ranged from 136.7–307.1 and was highest under sole FYM. The 50 % N_{FYM} recorded higher permanganate oxidisable carbon (POXC) and microbial biomass carbon (MBC) than 100 % RDF despite lower SOC, highlighting the importance of substrate quality. Enzyme activities were consistently lower in the cotton phase. Structural equation modelling indicated that SOC was associated with dehydrogenase activity indirectly through MBC, which mediated 86.5 % of this effect, whereas POXC was the primary driver of β -glucosidase (BGA) and cellulase activities (CA). These findings demonstrate that biological fertility in dryland Vertisols is governed by active, biologically accessible carbon pools rather than total SOC and that sustained organic and integrated inputs are essential for enhancing carbon quality and microbial enzyme function.

Keywords: carbon lability; carbon management index; enzyme activities; permanent manurial experiment; structural equation modelling

Introduction

Soil organic carbon (SOC) constitutes the largest active reservoir of terrestrial carbon and plays a pivotal role in regulating soil health, agricultural productivity and climate change mitigation. Its persistence is governed by the functional complexity of soil systems, arising from interactions among organic substrates, minerals, microorganisms and environmental conditions (1). In dryland and semi-arid agroecosystems, where limited precipitation and high evapotranspiration constrain biomass production and organic inputs, maintaining SOC stocks remains a major challenge, as they

are highly sensitive to nutrient management practices (2). Long-term studies consistently demonstrate that integration of organic manures with mineral fertilisers has been widely recognised as an effective strategy for enhancing SOC sequestration, improving soil quality and sustaining crop productivity (3, 4).

Vertisols, characterised by their high smectitic clay content and pronounced shrink-swell behaviour, are among the most agriculturally significant soils in peninsular India and other semi-arid tropical regions. Owing to their unique mineralogy, they possess a high capacity for carbon stabilisation through organo-mineral

associations and aggregate protection (1). Evidence from long-term fertilisation experiments indicates that integrated nutrient management (INM) significantly enhances SOC stocks and overall soil quality (4). Moreover, soil inorganic carbon (SIC), primarily present as pedogenic carbonates, represents a significant yet overlooked component of the total carbon pool in semi-arid environments, emphasising the need for comprehensive assessments of both organic and inorganic carbon fractions (5).

The SOC is inherently heterogeneous, comprising a continuum of fractions that differ in turnover time, biological accessibility and sensitivity to management. Operationally defined oxidisable C fractions including very labile (VLC), labile (LC), less labile (LLC) and non-labile (NLC) carbon, derived through graded $K_2Cr_2O_7$ oxidation provide insights into carbon stabilisation pathways under different management regimes (6). Among the active carbon indicators, permanganate oxidisable carbon (POXC) serves as a sensitive and rapid measure of management-induced changes in labile SOC and is strongly associated with microbial activity (7). Microbial biomass carbon (MBC), the living component of soil organic matter (SOM), is sensitive to substrate availability and management practices and is a key driver of nutrient cycling and carbon turnover (8). The carbon management index (CMI), which integrates carbon pool index (CPI) and lability index (LI), offers a holistic assessment of changes in both the quantity and quality of soil carbon relative to a reference system (3, 9).

Beyond the bulk SOC pool, the quality and availability of carbon fractions regulate microbial activity and the production of extracellular enzymes that drive SOM decomposition and nutrient cycling. Soil microorganisms and their enzymes are therefore key mediators of SOC transformation and stabilisation. Labile carbon pools such as MBC and POXC are closely linked with the activities of carbon-cycling enzymes, including β -glucosidase, dehydrogenase and cellulase, which catalyse the breakdown of plant-derived substrates and reflect soil biological fertility (10). Long-term fertilisation and residue incorporation enhance these enzyme activities and microbial abundance, reinforcing the tight coupling between SOC dynamics and biological processes under INM (11).

Despite substantial progress in understanding SOC dynamics, several knowledge gaps remain. Most previous studies have focused on individual carbon fractions or isolated biological indicators, whereas integrated multi-fraction assessments encompassing SOC, SIC, POXC, MBC, oxidisable carbon fractions, carbon quality indices and enzyme activities are scarce, particularly in long-term dryland Vertisol experiments in the Indian subcontinent. Furthermore, the mechanistic pathways linking SOC pools to specific enzyme responses are complex and often involve both direct and indirect interactions mediated by microbial communities. Conventional statistical approaches are limited in resolving these complex relationships, whereas structural equation modelling (SEM) provides a robust framework for disentangling causal pathways among soil carbon fractions, microbial biomass and enzyme activities (12).

The present study utilises a 43-year-old long-term field experiment, providing a rare opportunity to evaluate treatment-induced changes in soil carbon quality and microbial functionality that cannot be resolved in short-term studies. The prolonged and uninterrupted nutrient management history enables assessment of cumulative soil carbon dynamics and carbon sequestration under rainfed Vertisol conditions.

The specific objectives were to (i) evaluate the impact of long-term fertilisation and manuring on SOC and SIC pools in a rainfed maize-cotton Vertisol; (ii) characterise the distribution of oxidisable carbon fractions and assess carbon quality using carbon management indices; and (iii) elucidate the SOC-driven pathways to enzyme activities mediated by MBC and POXC using structural equation modelling (SEM).

Although multi-fraction assessments and SEM have been applied to soil-carbon studies, this study integrates oxidisable carbon fractionation, active carbon pools, carbon-quality indices and carbon-cycling enzyme activities within a 43-year dryland Vertisol experiment and uses SEM to resolve the specific mediating roles of MBC and POXC in linking bulk SOC to individual enzyme responses. This integrated framework provides mechanistic evidence linking long-term nutrient management to carbon quality and microbial functioning, thereby identifying the biologically active carbon pools that underpin enzyme activity under semi-arid rainfed conditions.

Materials and Methods

Experimental site

The study was conducted in a permanent manurial experiment (PME) established in 1982 at the Agricultural Research Station (ARS), Kovilpatti, Tamil Nadu Agricultural University (TNAU), India, one of the oldest continuously maintained dryland long-term experiments in peninsular India. The site experiences a hot semi-arid tropical climate; the long-term normals are a mean annual rainfall of 743 mm and a reference potential evapotranspiration (PET) of 812 mm, resulting in a water-deficit agro-ecosystem throughout most of the crop-growing period. Meteorological data for the study period were obtained from the on-site agro-meteorological observatory. Rainfall and evaporative demand differed markedly between the two crop years. The maize year (Rabi 2024–25) received 1188 mm of rainfall, whereas the cotton year (Rabi 2025–26) was considerably drier at 703 mm, with correspondingly higher Class-A pan evaporation (2008 and 2327 mm, respectively; Supplementary Table 1). As pan evaporation is not equivalent to PET, year-specific evaporative demand is reported as measured Class-A pan evaporation. The monthly rainfall distribution for the study period is presented in Supplementary Fig. 1. The soil is classified as Typic Haplusterts in USDA Soil Taxonomy, representing a deep, heavy-clay Vertisol characterised by high smectitic clay content, pronounced shrink-swell cracks during dry periods and strong self-mulching upon rewetting. The baseline soil characteristics of the experimental site recorded at the initiation of the PME in 1982, are presented in Supplementary Table 2. Over the course of the experiment, crops, varieties and associated agronomic practices were updated in accordance with regional recommendations, whereas the experimental layout and nutrient management treatments remained unchanged since 1982. Because the maize and cotton crops were grown and sampled in consecutive years (Rabi 2024–25 and 2025–26) rather than within a single year, the two cropping phases differ simultaneously in crop, sampling year and prevailing weather. Differences between the maize and cotton phases therefore cannot be attributed to a single factor and are interpreted throughout as combined crop-, year- and moisture-related effects rather than as effects of cropping period alone.

Experimental design and treatments

The experiment was conducted in a randomised block design (RBD) with 3 replications, with each plot measuring 7.5×3.6 m. All treatments have been maintained continuously since 1982 without interruption as part of the long-term PME. For the present study, maize (COHM8; 45×20 cm) was grown during rabi 2024–25, followed by cotton (KC3; 45×15 cm) during rabi 2025–26 under rainfed conditions. Nine nutrient management treatments were included in the experiment (Table 1). Fertiliser applications followed crop-specific recommended doses of fertilisers (RDF). All organic amendments were thoroughly incorporated into the topsoil during pre-sowing soil preparation each crop.

Soil sampling and processing

Surface soil samples (0–15 cm) were collected after the harvest of maize (January 2025) and cotton (March 2026) to capture nutrient management effects in each cropping phase. Using the quadrant sub-sampling approach, eight to ten randomly positioned cores per plot were composited to obtain one sample per replication. Field-moist subsamples (~50 g) were stored at 4 °C and analysed for microbial biomass carbon (MBC) within 48 hr. Remaining samples were air-dried under shade, gently disaggregated, sieved (< 2 mm) and stored for analyses. A portion of the < 2 mm soil was further ground to pass through a 0.5 mm sieve for the determination of oxidisable carbon fractions.

Soil analyses

The collected soil samples were analysed for physical, chemical and biological properties using standard laboratory procedures. The analytical methods employed for each parameter are described below.

Soil physical and chemical analysis

Soil moisture content at the time of sampling was determined gravimetrically by oven drying at 105 °C to constant weight and expressed on a dry weight basis. The moisture content was 32.7 % during maize sampling (January 2025) and 20.1 % during cotton sampling (March 2026). Initial physical, chemical and biological properties of the surface soil (0–15 cm), determined prior to the sowing of maize in rabi 2024–25, are summarised in Supplementary Table 3.

Soil carbon fractions

Soil carbon fractions and related properties were determined using standard analytical procedures as described. The SOC was determined by the wet oxidation (chromic acid digestion) method (13) of on soil ground to pass through a 0.5 mm sieve and expressed in g C kg^{-1} . Soil inorganic carbon (SIC) was determined using the acid neutralisation (rapid titration) method of (14). Results were calculated as CaCO_3 and expressed as g C kg^{-1} soil.

The MBC was determined by the chloroform fumigation-extraction method (15), as modified by previous researchers (16), where field-moist soil (10 g) was fumigated with ethanol-free chloroform for 24 hr alongside an unfumigated control and extracted with 0.5 M K_2SO_4 (1 : 4 w/v). The MBC was expressed as milligrams per kilogram (mg kg^{-1}).

The POXC was quantified using the 333 mM KMnO_4 oxidation method from previous reports (17), as modified and standardised by other researchers (7). Air-dried, 2 mm-sieved soil (2.5 g) was shaken with 18 mL of 333 mM KMnO_4 solution for 2 min and allowed to settle for 10 min. The supernatant was diluted 1 : 25 and absorbance read at 550 nm. Results were expressed as mg C kg^{-1} soil.

Oxidisable organic carbon fractions were determined following the modified Walkley-Black titrimetric procedure of (6). Briefly, soil samples were oxidised with 0.167 M $\text{K}_2\text{Cr}_2\text{O}_7$ in the presence of concentrated H_2SO_4 at three successive acid concentrations (12, 18 and 24 N) by adding 5, 10 and 20 mL of concentrated H_2SO_4 , respectively to 10 mL of $\text{K}_2\text{Cr}_2\text{O}_7$ solution. Excess dichromate was back titrated with 1.0 M FeSO_4 . Carbonates were removed by acid pretreatment prior to dry combustion and total organic carbon (TOC) was determined using a LECO elemental analyser (18). The procedure yielded four operationally defined fractions of decreasing oxidisability (i) VLC oxidised at 12 N H_2SO_4 ; (ii) LC the difference between carbon oxidised at 18 and 12 N H_2SO_4 ; (iii) LLC the difference between carbon oxidised at 24 and 18 N H_2SO_4 ; and (iv) NLC the residual carbon unoxidised even by 24 N H_2SO_4 . Each fraction was expressed in gram per kilogram (g kg^{-1}).

Soil enzyme activities

β -glucosidase activity (BGA) was determined using p-nitrophenyl- β -D-glucopyranoside (PNG) as the substrate following (19). About 1 g of soil was incubated with modified universal buffer (pH 6.0) at 37 °C for 1 hr. The reaction was terminated with CaCl_2 and NaOH and the released p-nitrophenol was measured at 400 nm. Results were expressed as $\text{mg p-nitrophenol (PNP) kg}^{-1} \text{ soil hr}^{-1}$. Dehydrogenase activity (DHA) was measured by the 2,3,5-triphenyltetrazolium chloride (TTC) reduction method of previous studies (20). Soil (6 g) was incubated with TTC and Tris-HCl buffer (pH 7.4) at 37 °C for 24 hr and the produced triphenyl formazan (TPF) was quantified at 485 nm, expressed as $\mu\text{g TPF g}^{-1} \text{ soil day}^{-1}$. Cellulase activity (CA) was determined by incubating 1 g of soil with acetate buffer (pH 5.5) containing 0.7 % carboxymethyl cellulose at 50 °C for 24 hr. Reducing sugars released during incubation were quantified spectrophotometrically at 540 nm and expressed as $\mu\text{g glucose equivalent g}^{-1} \text{ soil hr}^{-1}$ (21).

Table 1. Treatment details of the long-term nutrient management experiment under a rainfed maize-cotton cropping system on Vertisol at ARS Kovilpatti, Tamil Nadu, India

Treatments	Treatment details	Acronym
T1	Control (no input)	Control
T2	100 % RDF (Cotton: 40 : 20 : 40; Maize: 40 : 20 : 0 kg NPK ha^{-1})	100 % RDF
T3	50 % RDF (Cotton: 20 : 10 : 20; Maize: 20 : 10 : 0 kg NPK ha^{-1})	50 % RDF
T4	50 % nitrogen supplied through crop residues (CR)	50 % N_{CR}
T5	50 % nitrogen supplied through farmyard manure (FYM)	50 % N_{FYM}
T6	50 % inorganic N + 50 % organic N (CR) + 50 % P + 50 % K	INM_{CR}
T7	50 % inorganic N + 50 % organic N (FYM) + 50 % P + 50 % K	INM_{FYM}
T8	100 % RDF + 25 kg $\text{ZnSO}_4 \text{ ha}^{-1}$	RDF + Zn
T9	FYM (12.5 t ha^{-1})	FYM

RDF: recommended dose of fertilisers for the respective crop; FYM: farmyard manure; CR: crop residues; ZnSO_4 : zinc sulphate; INM: integrated nutrient management (combined application of 50 % inorganic N + 50 % organic N (through CR or FYM) + 50 % recommended P_2O_5 + 50 % recommended K_2O). All treatments were maintained continuously over the years under a rainfed, single crop per year system (Maize in rabi 2024–25 followed by cotton in rabi 2025–26).

Assessment of soil carbon indices

To evaluate the effects of long-term nutrient management on soil carbon quality and stability, CPI, LI and CMI were calculated as described in earlier reports (3, 9). In this study, POXC was used as the labile carbon fraction for index calculation, following the approach of previous studies as it represents a chemically defined, management-sensitive active carbon pool (7).

$$\text{Carbon lability (L)} = \frac{\text{Labile carbon (POXC)}}{\text{Non-labile carbon (SOC - POXC)}} \quad \text{Eq. (1)}$$

$$\text{Carbon pool index (CPI)} = \frac{\text{SOC}_{\text{treatment}}}{\text{SOC}_{\text{control}}} \quad \text{Eq. (2)}$$

$$\text{Lability index (LI)} = \frac{L_{\text{treatment}}}{L_{\text{control}}} \quad \text{Eq. (3)}$$

$$\text{Carbon management index (CMI)} = \text{CPI} \times \text{LI} \times 100 \quad \text{Eq. (4)}$$

Values of CPI and LI greater than 1 indicate increases in total SOC and carbon lability, respectively, while CMI values exceeding 100 denote improvement in soil carbon status relative to the reference soil (3, 9).

Long-term change in soil organic carbon and accumulation rate

Long-term changes in SOC were quantified by comparing current SOC with the baseline value of 1.5 g kg⁻¹ (mean SOC of the experimental plots) recorded at the initiation of the PME in 1982 using the previous method (13). The mean SOC of the maize (rabi 2024–25) and cotton (rabi 2025–26) was used to represent steady-state conditions. The absolute change in SOC (ΔSOC) and annual SOC accumulation rate was calculated as:

$$\Delta\text{SOC} = \text{SOC}_{\text{mean}} - \text{SOC}_{\text{baseline}} \quad \text{Eq. (5)}$$

$$\text{SOC accumulation rate} = \frac{\Delta\text{SOC}}{T} \quad \text{Eq. (6)}$$

where T is the duration of the experiment (43 years).

Statistical analysis

Data are presented as means of 3 replications. Before ANOVA, the normality of residuals (Shapiro-Wilk) and homogeneity of error variances (Levene's test) were verified; both assumptions were satisfied for all measured variables ($p > 0.05$), so untransformed data were analysed. Treatment effects were evaluated by ANOVA for the RBD, with treatment means separated by Fisher's protected least significant difference (LSD) test at $p \leq 0.05$ (22), applied only

after a significant treatment F-test. The choice of LSD was verified against the more conservative Tukey honestly significant difference (HSD) test, which left treatment rankings unchanged. The standard error of difference (SEd) and critical difference (CD) values are reported in the tables. All statistical analyses were performed in R version 4.3.2 (R Core Team, 2023). Pearson correlations, bivariate regressions and hierarchical clustering (Ward's method, Euclidean distance) of z-score standardised variables were performed (stats, ggplot2, corplot, factoextra). The SEM evaluated SOC-driven pathways to enzyme activities mediated by MBC and POXC (lavaan), with model fit assessed using standard goodness-of-fit indices (23). Given its low degrees of freedom ($df = 3$) and the pooling of non-independent observations ($n = 54$), the recursive SEM is treated as exploratory, inference rests on the standardised path coefficients and the principal SOC $\rightarrow$ MBC $\rightarrow$ dehydrogenase pathway was reproduced when each phase was fitted separately.

Results and Discussion

Soil organic carbon and inorganic carbon under long-term nutrient management

Long-term nutrient management significantly influenced all soil carbon pools across both crops (Table 2). The SOC ranged from 5.74 to 10.51 g kg⁻¹ (maize) and 5.66 to 11.19 g kg⁻¹ (cotton) across treatments. Integrated nutrient management using farmyard manure (INM_{FYM}) recorded the highest SOC in both phases (10.51 and 11.19 g kg⁻¹, respectively), followed closely by INM using crop residues (CR) (INM_{CR}) (10.35 and 10.78 g kg⁻¹) and farmyard manure (FYM) (10.32 and 10.63 g kg⁻¹). The minimum SOC was observed in the control (5.74 and 5.66 g kg⁻¹). This significantly higher SOC under INM_{FYM} and INM_{CR} compared to sole inorganic treatments highlights the important role of INM in enhancing organic carbon pools in rainfed Vertisols. These increases are attributed to sustained organic matter inputs that enrich both labile and stable carbon fractions (24), while improved root biomass and rhizodeposition under INM further contribute to below-ground carbon sequestration (8). The superior SOC under FYM and INM treatments is also linked to the biochemical composition of FYM, particularly lignin and polyphenols that promote humification and the formation of stable organo-mineral complexes with the smectite clays of Vertisols (25). Notably, SOC under sole FYM was comparable to INM, emphasising the importance of carbon input quality and quantity, consistent with global evidence on SOC enhancement through organic amendments (26). Although SOC increased under 100 % recommended dose of fertilisers (RDF) and

Table 2. Effect of long-term nutrient management on soil carbon fractions and related carbon pools under a rainfed maize-cotton cropping system

Treatment	SOC (g kg ⁻¹)		SIC (g kg ⁻¹)		MBC (mg kg ⁻¹)		POXC (mg kg ⁻¹)	
	Maize	Cotton	Maize	Cotton	Maize	Cotton	Maize	Cotton
Control	5.74 ± 0.11 ^f	5.66 ± 0.06 ^f	1.43 ± 0.03 ^h	1.17 ± 0.02 ^h	123 ± 1.17 ⁱ	104 ± 1.75 ^f	305 ± 1.24 ^h	281 ± 0.12 ^f
100 % RDF	9.03 ± 0.13 ^c	9.44 ± 0.16 ^c	4.00 ± 0.02 ^b	3.89 ± 0.04 ^b	174 ± 1.85 ^g	169 ± 1.58 ^e	628 ± 8.28 ^d	563 ± 1.68 ^c
50 % RDF	7.38 ± 0.13 ^e	7.7 ± 0.03 ^e	3.52 ± 0.02 ^d	3.27 ± 0.03 ^e	162 ± 1.04 ^h	168 ± 2.00 ^e	412 ± 2.63 ^g	384 ± 7.35 ^e
50 % N _{CR}	7.63 ± 0.07 ^e	8.22 ± 0.04 ^{de}	2.57 ± 0.05 ^g	2.34 ± 0.02 ^g	217 ± 3.05 ^e	206 ± 4.12 ^d	476 ± 9.11 ^f	451 ± 2.69 ^d
50 % N _{FYM}	8.06 ± 0.09 ^d	8.49 ± 0.03 ^d	2.76 ± 0.06 ^f	2.43 ± 0.03 ^g	258 ± 1.22 ^c	241 ± 4.31 ^c	593 ± 7.31 ^e	562 ± 3.11 ^c
INM _{CR}	10.35 ± 0.19 ^a	10.78 ± 0.01 ^{ab}	3.77 ± 0.05 ^c	3.44 ± 0.05 ^d	236 ± 4.52 ^d	239 ± 0.51 ^c	810 ± 9.31 ^c	727 ± 14.52 ^b
INM _{FYM}	10.51 ± 0.21 ^a	11.19 ± 0.07 ^a	3.96 ± 0.07 ^b	3.65 ± 0.04 ^c	286 ± 1.95 ^b	253 ± 2.48 ^b	843 ± 7.17 ^b	752 ± 14.38 ^{ab}
RDF + Zn	9.97 ± 0.11 ^b	10.25 ± 0.04 ^b	4.26 ± 0.06 ^a	4.12 ± 0.04 ^a	186 ± 0.4 ^f	172 ± 3.00 ^e	624 ± 11.41 ^d	583 ± 10.66 ^c
FYM	10.32 ± 0.18 ^a	10.63 ± 0.1 ^b	3.22 ± 0.07 ^e	2.83 ± 0.02 ^f	315 ± 5.76 ^a	307 ± 2.09 ^a	894 ± 12.54 ^a	773 ± 5.59 ^a
SEd	0.15	0.26	0.07	0.06	3.66	4.86	11.5	11.8
CD ($p < 0.05$)	0.33	0.54	0.15	0.12	7.77	10.29	24.5	25.1

Values represent means of replicated observations from surface soil (0–15 cm depth) sampled after crop harvest. SOC: soil organic carbon (g kg⁻¹); SIC: soil inorganic carbon (g kg⁻¹); MBC: microbial biomass carbon (mg kg⁻¹); POXC: permanganate oxidisable carbon (mg kg⁻¹). SEd: standard error of difference; CD: critical difference at 5 % level of significance. Treatments are described in Table 1.

RDF + zinc (Zn), it remained lower than under INM treatments, reflecting dependence on residue-derived carbon. The smectitic clay mineralogy of Vertisols further stabilises SOC through aggregate formation and clay-humus complexes, protecting it from decomposition in semi-arid environments (4). The higher SOC during the cotton season is associated with lignified residues and deeper root systems, reinforcing the influence of cropping systems on SOC accumulation (27).

The SIC followed a markedly different treatment ranking from SOC. The RDF + Zn had the highest SIC (4.26 and 4.12 g kg⁻¹ for maize and cotton, respectively) and 100 % RDF (4.00 and 3.89 g kg⁻¹). The control exhibited the lowest SIC (1.43 and 1.17 g kg⁻¹), while 50 % N_{FYM} (2.76 and 2.43 g kg⁻¹) and 50 % N_{CR} (2.57 and 2.34 g kg⁻¹) recorded lower SIC relative to inorganic-dominant treatments despite higher SOC, indicating a divergence between organic and inorganic carbon accumulation pathways. While SOC responds primarily to organic inputs, SIC is largely governed by soil chemistry and climatic conditions that favour carbonate formation under semi-arid environments (28). Greater SIC under inorganic fertilisation may be associated with alkaline conditions and increased availability of Ca²⁺ and Mg²⁺, whereas enhanced microbial activity under organic amendments promotes carbonate dissolution through CO₂ and organic acid production (29). Similar management responses of SIC have been reported in dryland Vertisols (30). The relatively small seasonal variation further suggests that SIC is influenced more by inherent soil properties than by short-term biological inputs (27), highlighting the complementary roles of SOC and SIC in evaluating carbon sequestration potential (31).

Active soil carbon pools under rainfed permanent manurial experiment

Across treatments, MBC varied from 123 to 315 mg kg⁻¹ in the maize phase and from 104 to 307 mg kg⁻¹ in the cotton phase. The application of FYM recorded the highest MBC in both crops (315 and 307 mg kg⁻¹), followed by INM_{FYM} (286 and 253 mg kg⁻¹). The control had the lowest MBC (123 and 104 mg kg⁻¹). The MBC, an active fraction of SOM, was thus highest under FYM and INM_{FYM} treatments, demonstrating the importance of organic inputs in sustaining microbial populations. Long-term fertilisation studies consistently demonstrate that the incorporation of FYM, either alone or in combination with mineral fertilisers, significantly enhances MBC compared with sole inorganic fertilisation (8, 32). Notably, 50 % N_{FYM} (258 and 241 mg kg⁻¹) exceeded 100 % RDF (174 and 169 mg kg⁻¹) despite having comparatively lower SOC, emphasising the dominant role of substrate quality and decomposability over total carbon stock (33). Vertisols inherently support higher microbial biomass due to their smectitic clay mineralogy and superior moisture retention (4) and continuous inputs of labile carbon under FYM and INM further stimulate microbial activity (10, 26), while FYM-based systems consistently outperform chemical fertilisation in enhancing microbial biomass (34). The lower MBC was observed during the cotton phase, which is consistent with the lower soil moisture at sampling (20.1 % vs 32.7 %). However, because the maize and cotton phases also differed in crop, sampling year and prevailing weather, this decline cannot be attributed to moisture alone and may additionally reflect differences in CR and inter-annual climatic variation. As a sensitive indicator of soil quality, MBC showed a strong positive association with SOC, emphasising its central role in SOM stabilisation (8).

The POXC ranged from 281 to 894 mg kg⁻¹ across phases and treatments and was the most treatment responsive carbon fraction. The FYM recorded the highest POXC in both phases (894 and 773 mg kg⁻¹). The lowest POXC was observed in the control (305 and 281 mg kg⁻¹). Among inorganic treatments, 100 % RDF (628 and 563 mg kg⁻¹) and RDF + Zn (624 and 583 mg kg⁻¹) exceeded 50 % RDF (412 and 384 mg kg⁻¹). The highest levels under FYM and INM_{FYM} highlight the role of organic inputs in enriching biologically available carbon, as FYM provides readily decomposable substrates, including carbohydrates and microbial metabolites, that stimulate microbial activity and enhance labile carbon pools; a similar increase in POXC with organic amendments has been widely documented (35). Although balanced inorganic fertilisation indirectly enhances POXC through increased biomass production and rhizodeposition, its effect remains inferior to organic treatments due to the absence of immediate organic substrates. The POXC is closely associated with processed particulate organic matter and is highly sensitive to management changes (7). Furthermore, previous studies emphasised that POXC represents a chemically defined carbon fraction sensitive to lignin-rich organic matter rather than strictly labile carbon, indicating that elevated POXC under FYM in this Vertisol reflects both readily decomposable substrates and ligno-humic precursor accumulation (36). Across all treatments, POXC declined from the maize to cotton, paralleling the seasonal trend in MBC.

Long-term change in soil organic carbon and accumulation rate

Relative to the baseline SOC of 1.5 g kg⁻¹ recorded in 1982 (13), SOC increased under all treatments over the 43-year experimental period (Fig. 1). The increase (Δ SOC) ranged from 4.20 g kg⁻¹ in the control to 9.35 g kg⁻¹ under INM_{FYM}, corresponding to annual SOC accumulation rates of 0.098–0.217 g kg⁻¹ yr⁻¹. The largest gains were recorded under INM_{FYM} (9.35 g kg⁻¹; 0.217 g kg⁻¹ yr⁻¹), followed by INM_{CR} (9.06 g kg⁻¹; 0.211 g kg⁻¹ yr⁻¹) and FYM (8.98 g kg⁻¹; 0.209 g kg⁻¹ yr⁻¹). In comparison, 100 % RDF resulted in a moderate increase (7.73 g kg⁻¹; 0.180 g kg⁻¹ yr⁻¹), whereas the control showed the smallest gain (4.20 g kg⁻¹; 0.098 g kg⁻¹ yr⁻¹). The consistently greater SOC accumulation under FYM- and INM-based treatments demonstrates the long-term benefit of organic inputs in enhancing soil carbon sequestration compared with sole inorganic fertilisation. These findings are consistent with long-term studies reporting progressive SOC build-up under organic and integrated nutrient management in Vertisols (4, 24).

Oxidisable organic carbon fractions and their distribution

The graded K₂Cr₂O₇ oxidation procedure resolved SOC into four operationally defined fractions including VLC, LC, LLC and NLC, whose absolute concentrations (g kg⁻¹) across treatments are presented in Fig. 2.

Very labile ranged from 0.96 to 2.46 g kg⁻¹ (maize) and 0.65 to 2.39 g kg⁻¹ (cotton). The highest VLC in both phases was recorded in FYM treatment (2.46 and 2.39 g kg⁻¹), followed by 50 % N_{FYM} (2.24 and 2.11 g kg⁻¹) and INM_{FYM} (2.17 and 2.04 g kg⁻¹). The control had the lowest VLC (0.96 and 0.65 g kg⁻¹). This dominance under FYM and 50 % N_{FYM} reflects the capacity of FYM to supply readily available labile substrates for microbial decomposers. The LC varied from 0.81 to 2.81 g kg⁻¹ across phases and treatments. In contrast to VLC, INM treatments exhibited higher LC than sole FYM. The INM_{FYM} showed the highest LC in both phases (2.44 and 2.81 g kg⁻¹), followed by INM_{CR} (2.31 and 2.46 g kg⁻¹) and FYM (2.24 and 2.27 g kg⁻¹). This indicates that INM supplies a continuous and varied carbon source, enriching the

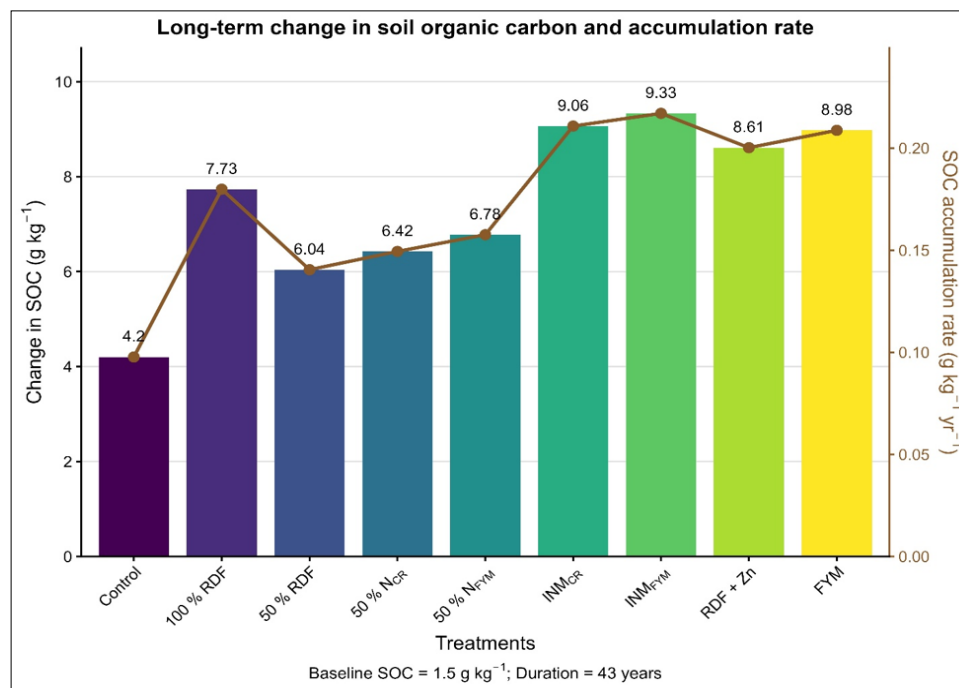


Fig. 1. Changes in soil organic carbon (SOC) and annual accumulation rates under long-term nutrient management in a rainfed semi-arid Vertisol.

RDF: recommended dose of fertilisers; N_{cr}: no crop residue; N_{fym}: no farmyard manure; INM_{cr}: integrated nutrient management with crop residue; INM_{fym}: integrated nutrient management with farmyard manure; FYM: farmyard manure; Zn: zinc; SOC: soil organic carbon. Bars denote ΔSOC (g kg^{-1}) relative to the 1982 baseline (1.5 g kg^{-1}) and the line represents SOC accumulation rates ($\text{g kg}^{-1} \text{yr}^{-1}$) over 43 years. Treatments are described in Table 1.

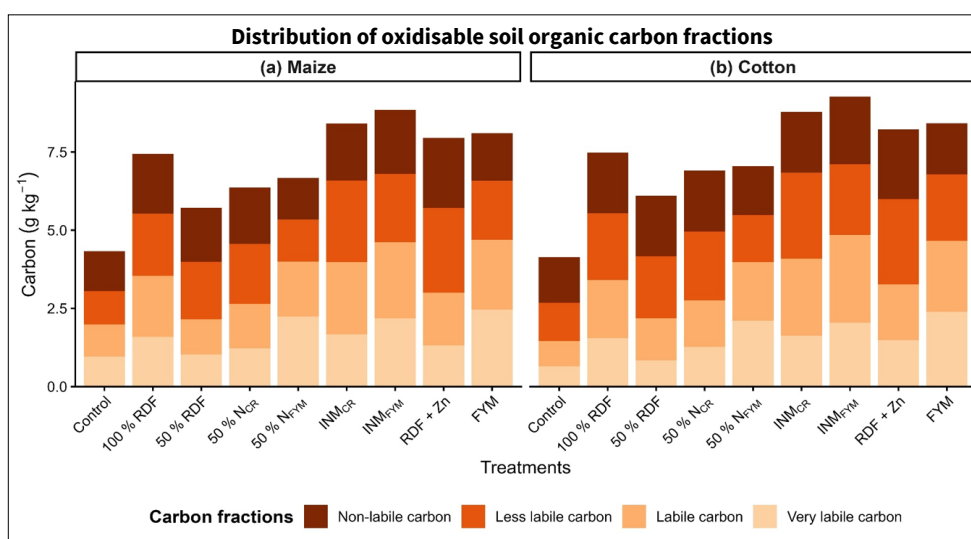


Fig. 2. Distribution of oxidisable soil organic carbon fractions (very labile, labile, less labile and non-labile carbon) under different nutrient management treatments. a: maize; b: cotton phases.

RDF: recommended dose of fertilisers; N_{cr}: no crop residue; N_{fym}: no farmyard manure; INM_{cr}: integrated nutrient management with crop residue; INM_{fym}: integrated nutrient management with farmyard manure; FYM: farmyard manure; Zn: zinc; SOC: soil organic carbon. Bars represent the cumulative contribution of each fraction expressed in g kg^{-1} soil. Carbon fractions are arranged from most stable (top) to most labile (bottom).

intermediate SOC fractions through the additive effects of organic and inorganic inputs. Previous researchers found that combined manure and fertiliser inputs act complementarily on labile fractions, so that no single input type improves all aspects of carbon quality at once (37). A comparable enrichment of VLC and LC under organic nutrient management in Vertisols was reported by previous researchers, who linked it to a shift of SOC toward biologically active pools (38).

The LLC was highest under RDF + Zn (2.71 and 2.78 g kg^{-1} for maize and cotton), INM_{cr} (2.61 and 2.75 g kg^{-1}) and lowest under the control (1.07 and 1.20 g kg^{-1}). The RDF + Zn (2.24 and 2.27 g kg^{-1}) recorded the highest NLC, followed by 50 % RDF (1.73 and 1.81 g kg^{-1})

and lowest in the control (1.27 and 1.46 g kg^{-1}) and FYM (1.56 and 1.65 g kg^{-1}). The greater LLC and NLC under RDF + Zn indicate increased stabilisation of carbon into relatively resistant pools, likely through enhanced biomass production together with aggregate protection and mineral associations in the clay-rich Vertisols. Zinc supplementation may also contribute indirectly by modifying rhizosphere microbial processes that influence below-ground carbon inputs (39). In contrast, the comparatively lower NLC under FYM despite higher SOC reflects the allocation of a larger proportion of carbon to biologically active fractions rather than recalcitrant pools, consistent with the carbon lability concept proposed by earlier researchers (7).

Carbon management index, carbon pool index and lability index

The CMI, CPI and LI calculated using POXC as the labile fraction, are presented in Supplementary Table 4 and illustrated in Fig. 3. As an index that integrates CPI and LI relative to a reference treatment, CMI provides an integrative measure of management-induced change in soil carbon quantity and lability (3, 9).

The CMI ranged from 136.7 (50 % RDF, cotton) to 307.1 (FYM, maize), with all fertilised treatments exceeding the control (100.0), confirming that fertilisation improved carbon quality relative to the unfertilised soil. The FYM recorded the highest CMI in both phases (307.1 maize; 281.9 cotton), followed by INM_{FYM} (287.5; 272.6) and INM_{CR} (275.7; 263.7). The overall hierarchy (FYM > INM_{FYM} > INM_{CR} > 100 % RDF > RDF + Zn > 50 % N_{FYM} > 50 % N_{CR} > 50 % RDF) demonstrates that FYM- and INM-based management most effectively enhanced both carbon accumulation and lability, consistent with the improved CMI, stratification ratio and sustainable yield index reported under long-term organic and integrated nutrient management (25). In contrast, the consistently low CMI under 50 % RDF (136.9 maize; 136.7 cotton) indicates that reduced inorganic fertilisation alone is insufficient to enrich biologically active carbon pools, corroborating observations from other long-term semi-arid experiments (3). The intermediate CMI under RDF + Zn (208.8 maize; 209.1 cotton), suggests that zinc supplementation favoured carbon stabilisation rather than enrichment of labile carbon fractions.

The CPI ranged from 1.286 (50 % RDF, maize) to 1.977 (INM_{FYM}, cotton), with generally higher values during the cotton phase. INM_{FYM} recorded the highest CPI, followed by INM_{CR} and FYM. In contrast, LI declined from the maize to the cotton phase across all treatments and was highest under FYM (1.708 and 1.501) and lowest under 50 % RDF (1.065 and 1.005). Despite the increase in CPI, the concurrent decline in LI and CMI indicates that the accumulation of total SOC during the cotton phase was accompanied by a proportional reduction in the biologically active carbon pool. Because the maize and cotton phases differed simultaneously in crop, sampling year and weather, this pattern reflects their combined effects rather than moisture alone. Nevertheless, the lower soil moisture during the cotton phase likely reduced microbial activity and labile carbon turnover, resulting in a greater proportion of carbon in less labile forms. These findings emphasize the importance of evaluating both carbon quantity and quality through CMI for assessing long-term soil carbon dynamics, consistent with observations in dryland agroecosystems (40).

Soil enzyme activities as indicators of carbon-driven biological fertility

The carbon-cycling enzymes responded strongly to long-term nutrient management, underlining their value as indicators of biologically mediated fertility governed by carbon availability in Vertisols (Table 3). The BGA activity ranged from 48.5 to 84.0 mg PNP kg⁻¹ hr⁻¹ (maize) and 42.6 to 76.4 mg PNP kg⁻¹ hr⁻¹ (cotton). INM_{FYM} recorded the highest BGA in both phases (84.0 and 76.4 mg PNP kg⁻¹ hr⁻¹), followed by RDF + Zn (80.7 and 72.7 mg PNP kg⁻¹ hr⁻¹) and INM_{CR} (78.6 and 70.2 mg PNP kg⁻¹ hr⁻¹). The control had the lowest BGA activity in both phases (48.5 and 42.6 mg PNP kg⁻¹ hr⁻¹). FYM (74.3 and 67.5 mg PNP kg⁻¹ hr⁻¹) ranked below RDF + Zn for BGA despite its higher MBC and POXC. The greater BGA under INM_{FYM} reflects the synergistic effects of organic inputs and balanced

fertilisation in enhancing SOC and labile carbon (POXC), thereby supporting microbial growth and extracellular enzyme production (35, 41, 42). Although FYM maintained higher MBC and POXC, its lower BGA than RDF + Zn suggests that BGA expression depends not only on substrate availability but also on nutrient stoichiometry and microbial functional dynamics (41).

The DHA ranged from 17.5 to 45.0 µg TPF g⁻¹ day⁻¹ across phases. The FYM recorded the highest DHA (45.0 maize and 39.8 cotton µg TPF g⁻¹ day⁻¹), followed by INM_{FYM} (42.6 and 37.5 µg TPF g⁻¹ day⁻¹), whereas lowest was observed in the control (20.2 and 17.5 µg TPF g⁻¹ day⁻¹). The greater DHA under FYM is consistent with its higher SOC and MBC, supporting the close association between microbial metabolic activity and carbon availability. Similar positive relationships between SOC, MBC and DHA under long-term organic and integrated nutrient management have been reported by earlier researchers (43). These contrasting responses suggest functional differentiation, with INM_{FYM} favouring hydrolytic enzyme activity and FYM supporting microbial metabolic activity.

The cellulase activity (CA) ranged from 16.2 to 35.2 µg glucose g⁻¹ hr⁻¹ across treatments and phases. The INM_{FYM} recorded the highest CA (35.2 maize; 31.6 cotton µg glucose g⁻¹ hr⁻¹), followed by RDF + Zn (33.4 and 29.8 µg glucose g⁻¹ hr⁻¹). The FYM (29.6 and 26.7 µg glucose g⁻¹ hr⁻¹) ranked below both INM and RDF + Zn for CA, consistent with the LC distribution. The control had the lowest CA (18.6 and 16.2 µg glucose g⁻¹ hr⁻¹). The higher CA under INM_{FYM} indicates that integrated nutrient management promotes the decomposition of cellulose and structurally complex carbon substrates through the combined supply of organic substrates and balanced nutrients (41). The closer correspondence of CA with LC and LLC than with VLC or MBC further supports its role in degrading partially processed carbon pools, consistent with findings of previous reports that combined mineral fertilisation and residue incorporation enhance CA through enrichment of labile carbon pools such as POXC (11). The comparatively lower CA under sole FYM suggests that optimum cellulase expression requires not only abundant carbon substrates but also a balanced nutrient supply, including N, P and Zn.

The relatively high BGA and CA under RDF + Zn further indicate the importance of balanced nutrient supply and zinc as an enzymatic cofactor supporting microbial metabolism and enzyme activity (39), whereas the consistently lower activities in the control reflect limited organic substrate availability (44). All three enzyme activities declined from the maize to the cotton phase. This decline was consistent with the lower soil moisture at cotton sampling (20.1 % vs 32.7 %). However, as the two phases also differed in crop, sampling year and prevailing weather, the observed reduction cannot be attributed to soil moisture alone and may additionally reflect differences in CR and inter-annual climatic variation (45).

Relationships among soil carbon fractions and enzyme activities

The Pearson correlation matrix (Fig. 4) resolved the measured variables into two major co-variation clusters. The first, comprising VLC, MBC and DHA, which showed extremely strong positive correlations ($R = 0.92\text{--}0.99$), corresponds to the biologically active, labile end of the soil carbon continuum and is highly responsive to organic management. The second cluster consisted of SOC, BGA and CA ($R = 0.93\text{--}0.99$), representing a more integrated carbon pool associated with hydrolytic enzyme activity, with POXC and LC bridging the two clusters. In contrast, the passive fractions (SIC, LLC

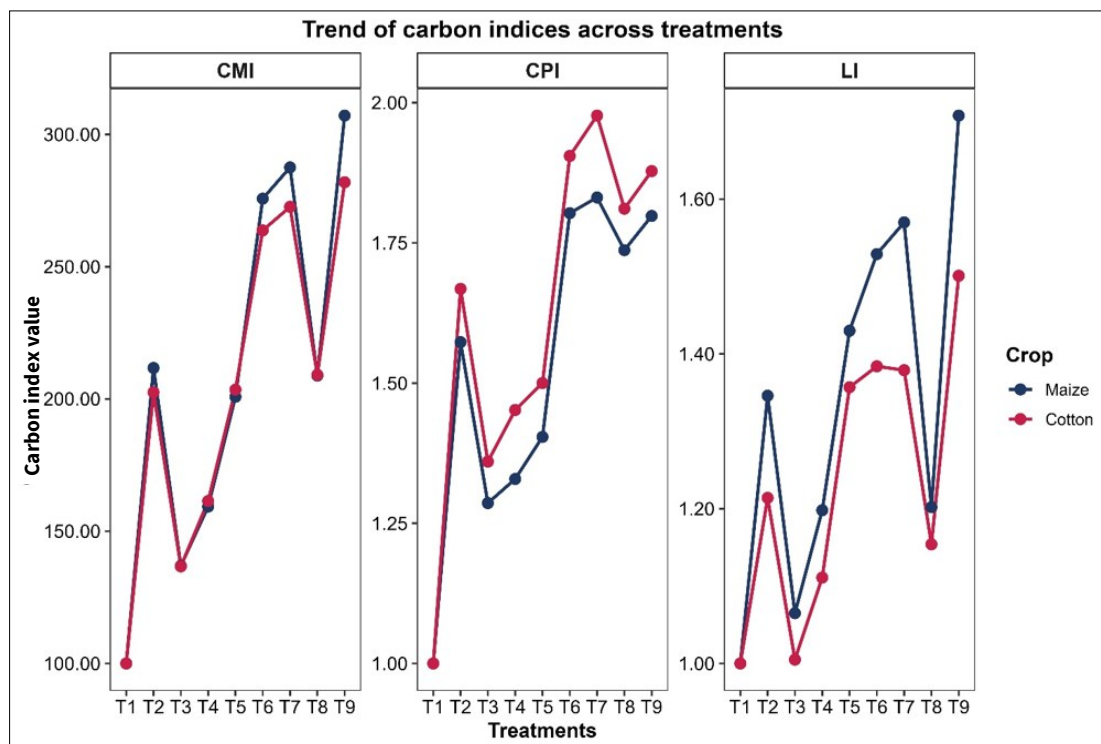


Fig. 3. Trend of carbon management index (CMI), carbon pool index (CPI) and lability index (LI) across nine treatments (T1-T9) for maize and cotton crops. Lines represent mean carbon index values for each crop, illustrating treatment-wise variation in soil carbon quality indicators. Treatments are described in Table 1.

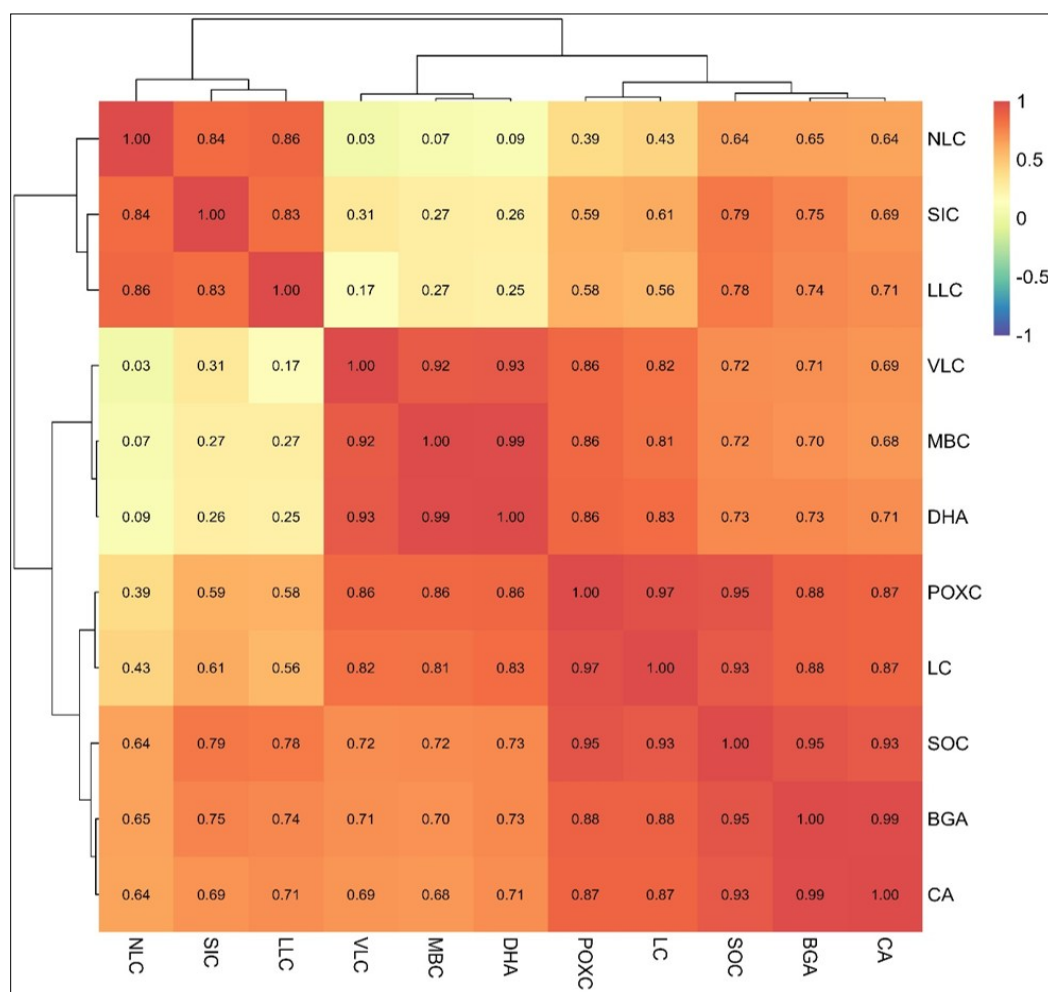


Fig. 4. Pearson correlation matrix showing pairwise relationships among soil carbon fractions and enzyme activities under long-term nutrient management treatments. Hierarchical clustering dendrograms indicate groupings of positively and negatively correlated variables. Colour scale ranges from deep red ($r = 1$) to blue ($r = -1$). SOC: soil organic carbon; SIC: soil inorganic carbon; MBC: microbial biomass carbon; POXC: permanganate oxidisable carbon; VLC: very labile carbon; LC: labile carbon; LLC: less labile carbon; NLC: non-labile carbon; BGA: β -glucosidase activity; DHA: dehydrogenase activity; CA: cellulase activity.

Table 3. Effect of long-term nutrient management on soil enzyme activities under a rainfed maize-cotton cropping system on Vertisol

Treatments	Beta glucosidase activity (mg PNP kg ⁻¹ hr ⁻¹)		Dehydrogenase activity (μg TPF g ⁻¹ day ⁻¹)		Cellulase activity (μg glucose g ⁻¹ soil hr ⁻¹)	
	Maize	Cotton	Maize	Cotton	Maize	Cotton
Control	48.5 ± 0.79 ^h	42.6 ± 0.34 ^h	20.2 ± 0.28 ^h	17.5 ± 0.34 ^h	18.6 ± 0.31 ⁱ	16.2 ± 0.14 ^h
100 % RDF	65.2 ± 0.73 ^e	59.5 ± 0.23 ^e	27.3 ± 0.21 ^f	23.9 ± 0.25 ^f	24.8 ± 0.34 ^f	22.1 ± 0.19 ^f
50 % RDF	56.6 ± 0.51 ^g	49.3 ± 0.55 ^g	24.8 ± 0.47 ^g	21.6 ± 0.14 ^g	21.5 ± 0.44 ^h	19.3 ± 0.08 ^g
50 % N _{CR}	62.4 ± 0.85 ^f	55.8 ± 0.01 ^f	32.5 ± 0.05 ^e	28.7 ± 0.36 ^e	23.9 ± 0.14 ^g	21.4 ± 0.45 ^f
50 % N _{FYM}	70.2 ± 1.2 ^d	63.6 ± 0.76 ^d	38.4 ± 0.63 ^c	33.9 ± 0.35 ^c	27.6 ± 0.56 ^e	24.8 ± 0.51 ^e
INM _{CR}	78.6 ± 0.41 ^b	70.2 ± 1.47 ^{bc}	35.8 ± 0.75 ^d	31.6 ± 0.32 ^d	31.8 ± 0.02 ^c	28.5 ± 0.34 ^c
INM _{FYM}	84.0 ± 0.65 ^a	76.4 ± 0.39 ^a	42.6 ± 0.39 ^b	37.5 ± 0.64 ^b	35.2 ± 0.54 ^a	31.6 ± 0.42 ^a
RDF + Zn	80.7 ± 0.14 ^b	72.7 ± 1.06 ^b	28.6 ± 0.2 ^f	25.1 ± 0.53 ^f	33.4 ± 0.57 ^b	29.8 ± 0.63 ^b
FYM	74.3 ± 1.24 ^c	67.5 ± 0.58 ^c	45.0 ± 0.62 ^a	39.8 ± 0.52 ^a	29.6 ± 0.12 ^d	26.7 ± 0.21 ^d
SEd	1.07	1.64	0.68	0.63	0.41	0.59
CD (<i>p</i> < 0.05)	2.26	3.48	1.44	1.34	0.88	1.25

Values represent mean of replicated observations. β-glucosidase activity as mg PNP (p-nitrophenol) kg⁻¹ soil hr⁻¹; dehydrogenase activity as μg TPF (triphenyl formazan) g⁻¹ day⁻¹; Cellulase activity is expressed as μg glucose g⁻¹ soil hr⁻¹. SEd: standard error of difference; CD: critical difference at *p* < 0.05. Treatment details are given in Table 1.

and NLC) associated only weakly with MBC, DHA and VLC (*R* = 0.03–0.31), consistent with their greater stability and the limited contribution of secondary carbonates and recalcitrant carbon to microbial metabolism. Bivariate regression analyses confirmed strong positive relationships between the active carbon pools (SOC, MBC and POXC) and enzyme activities in both the maize (Fig. 5) and cotton (Fig. 6) phases. The POXC exhibited stronger associations

with enzyme activities than SOC, while MBC showed the strongest relationship with DHA (*R* = 0.92–0.94), reflecting the close coupling between microbial biomass and oxidative enzyme activity. SOC also showed strong positive correlations with BGA (*R* = 0.95), CA (*R* = 0.93), POXC (*R* = 0.95), MBC (*R* = 0.72) and DHA (*R* = 0.73), highlighting its close association with soil biological activity. The stronger association of reactive, processable carbon with enzyme

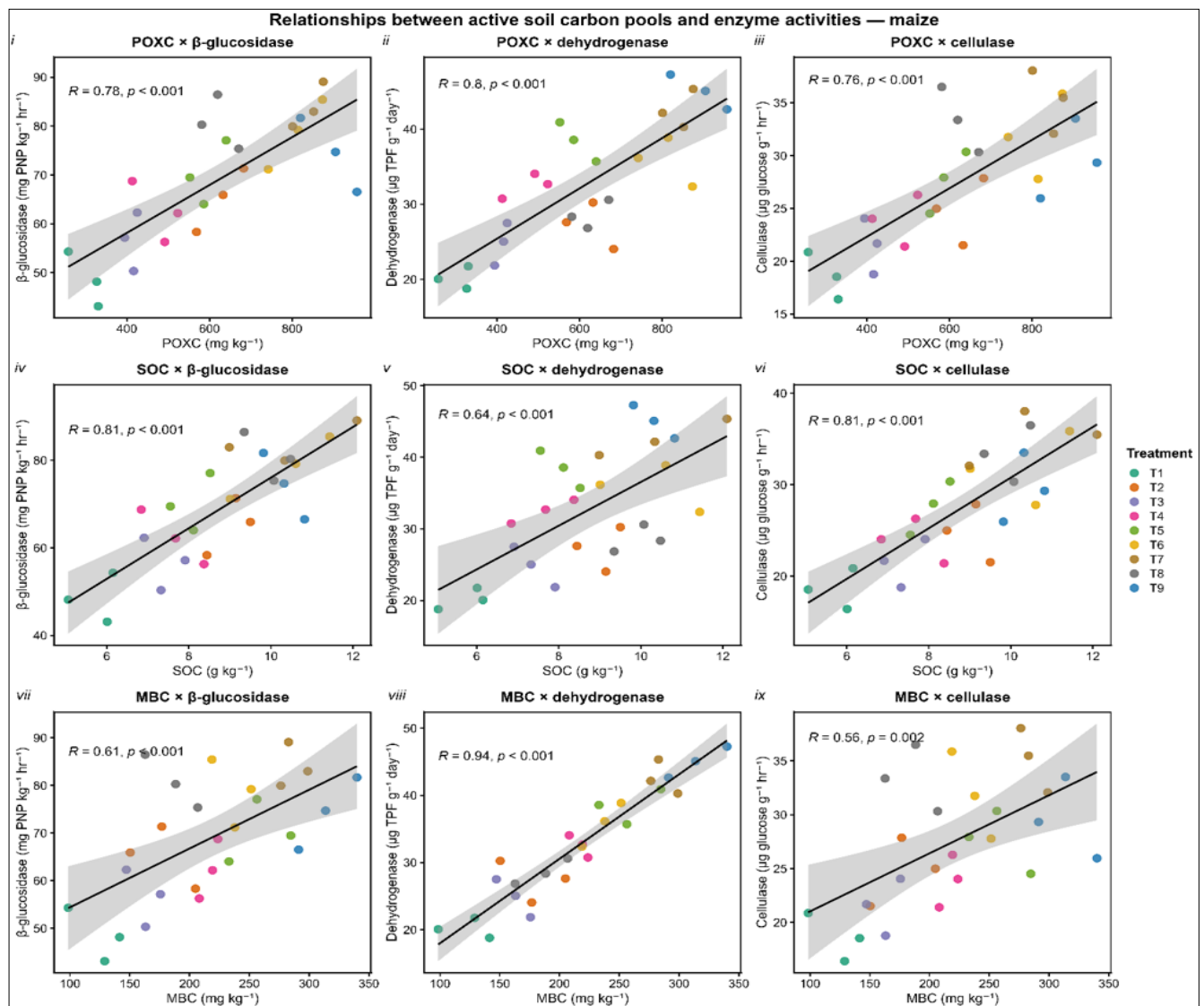


Fig. 5. Scatter plots depicting relationships between active soil carbon pools [permanganate oxidisable carbon (POXC), soil organic carbon (SOC) and microbial biomass carbon (MBC)] and enzyme activities (β-glucosidase, dehydrogenase and cellulase) during the maize phase. Each point represents an individual replicate (*n* = 27; 9 treatments × 3 replications) coloured by treatment (T1–T9). Pearson correlation coefficients (*R*) and significance levels are shown for each relationship. Solid lines represent linear regression fits; shaded areas indicate 95 % confidence intervals. Treatments are described in Table 1.

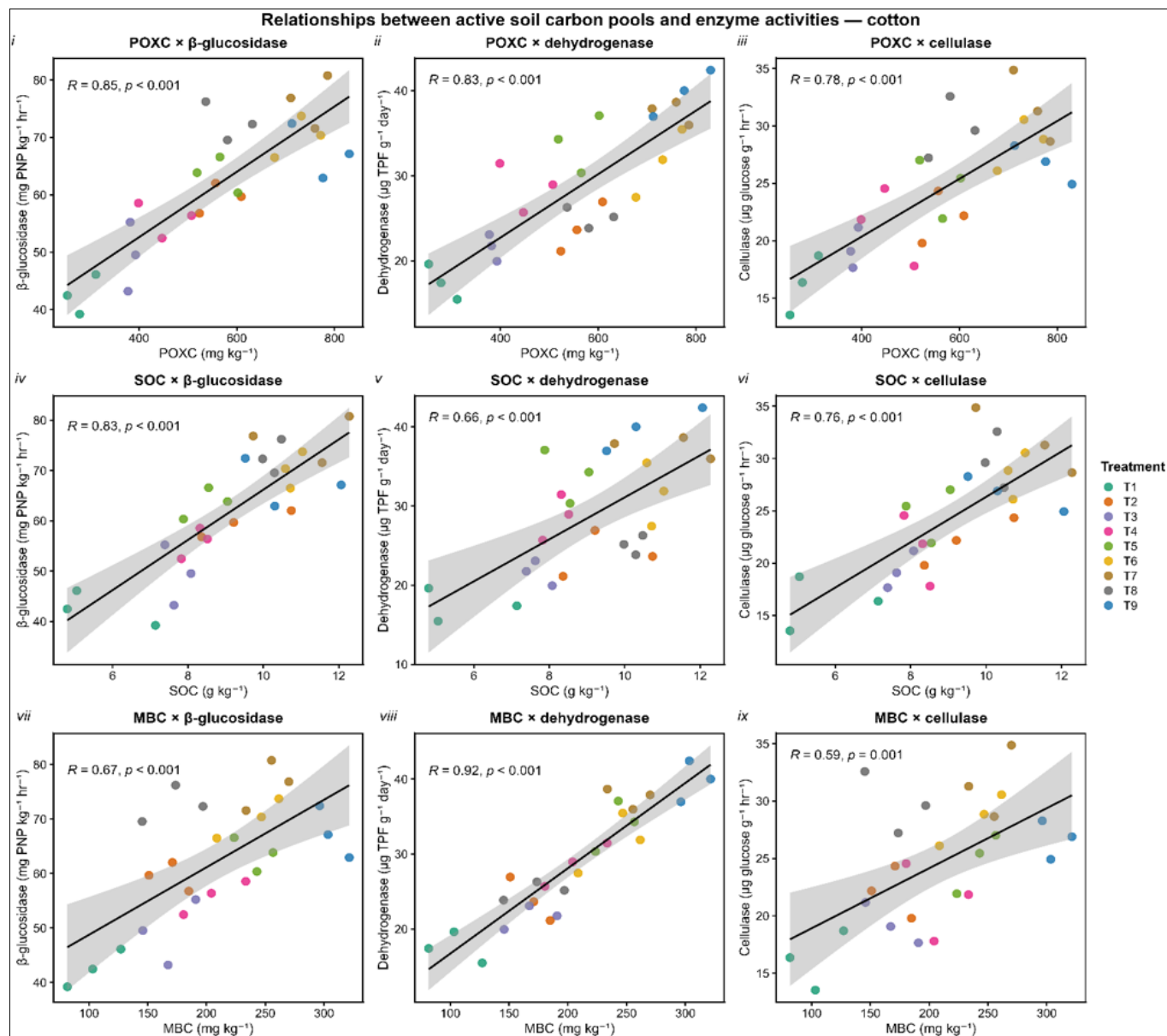


Fig. 6. Scatter plots depicting relationships between active soil carbon pools [permanganate oxidisable carbon (POXC), soil organic carbon (SOC) and microbial biomass carbon (MBC)] and enzyme activities (β-glucosidase, dehydrogenase and cellulase) during the cotton phase. Each point represents an individual replicate (n = 27; 9 treatments × 3 replications) coloured by treatment (T1–T9). Pearson correlation coefficients (R) and significance levels are shown for each relationship. Solid lines represent linear regression fits; shaded areas indicate 95 % confidence intervals. Treatments are described in Table 1.

activity is consistent with the role of POXC as a readily available microbial substrate and a more responsive indicator of biological soil function than total SOC (7). Similarly, other studies reported that carbon-cycle enzymes are more closely associated with labile carbon fractions than with total SOC under long-term nutrient management (11). In rainfed Vertisols, where crop, sampling year and prevailing weather collectively influence microbial activity, the close POXC-enzyme association represents an important indicator of soil carbon quality.

Oxidisable carbon fractions exhibited distinct relationships with enzyme activities (Fig. 7 and 8). The VLC and LC were strongly and positively associated with enzyme activities, whereas LLC showed weaker relationships and NLC the weakest, reflecting progressively lower biological relevance with increasing carbon recalcitrance. The strong association between VLC and DHA indicates the dependence of microbial metabolism on readily available carbon substrates, while the closer relationship between LC and CA is consistent with the role of cellulase in decomposing partially processed cellulosic materials represented within the LC

fraction. These substrate-enzyme relationships support the concept of ecological stoichiometry proposed by previous researchers, whereby microbial communities allocate enzyme production in response to the most limiting but biochemically accessible carbon substrates (46).

Causal pathways from soil organic carbon to enzyme activity: structural equation modelling

The full structural equation model (SEM) showed an excellent fit to the observed data (Table 4), indicating that the hypothesised relationships adequately represented the underlying soil carbon-enzyme dynamics. The chi-square test was non-significant ($\chi^2 = 2.982, p = 0.394$), indicating no significant discrepancy between the model-implied and observed covariance matrices. Comparative fit index (CFI), Tucker-Lewis index (TLI) were both well above the 0.95 threshold. Root mean square error of approximation (RMSEA) was 0.002 (0.000–0.229) and standardised root mean square residual (SRMR) was 0.011, both within acceptable limits. The wide upper RMSEA confidence interval reflects the low degrees of freedom

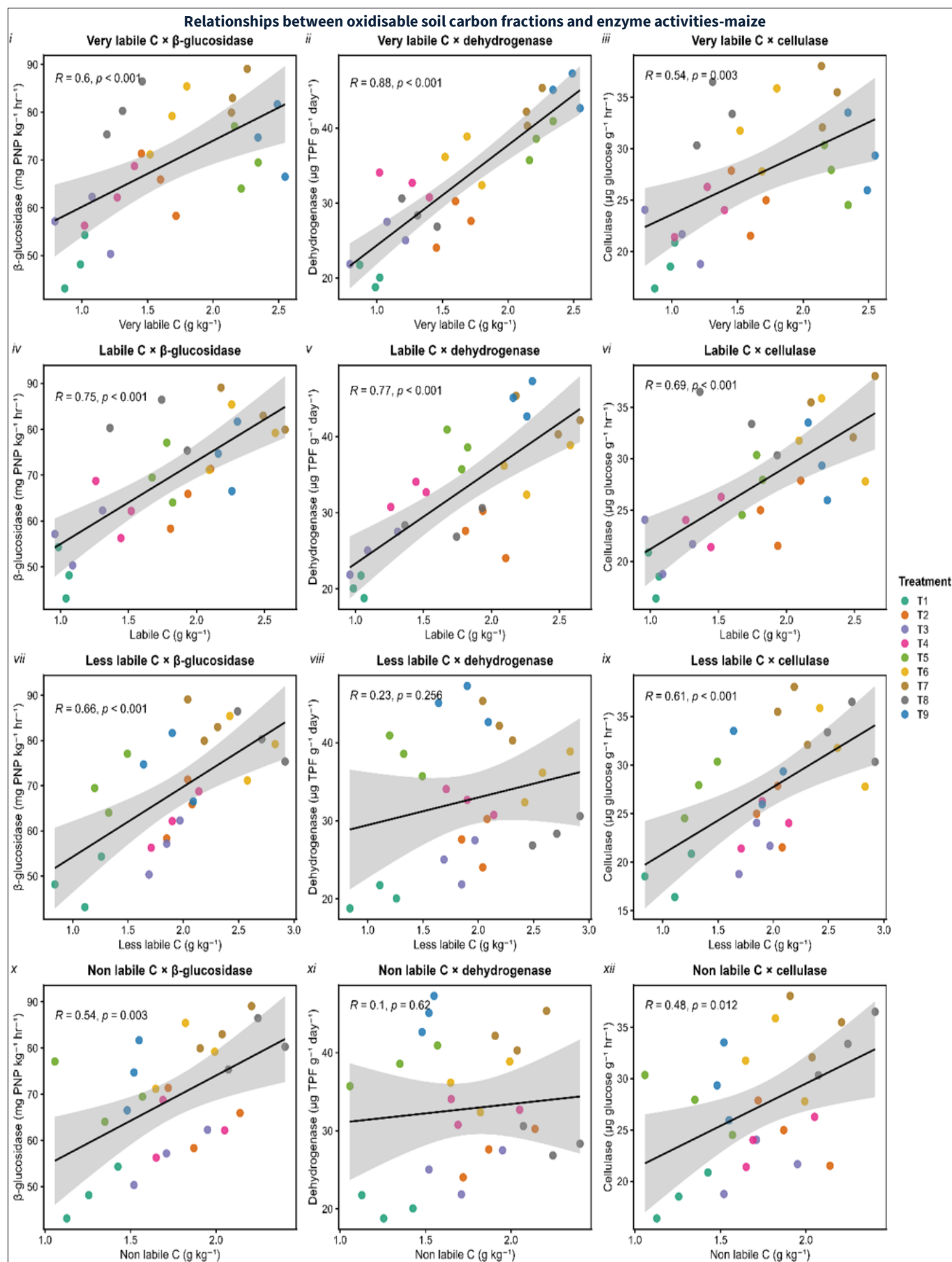


Fig. 7. Scatter plots illustrating relationships between oxidisable soil carbon fractions (very labile C, labile C, less labile C and non-labile C) and enzyme activities (β -glucosidase, dehydrogenase and cellulase) during the maize phase. Each point represents an individual replicate ($n = 27$; 9 treatments $\times$ 3 replications), colour-coded by treatment (T1-T9), with Pearson correlation coefficients (R) and p -values indicated for each panel. Solid lines represent linear regression fits; shaded areas indicate 95 % confidence intervals. Treatments are described in Table 1.

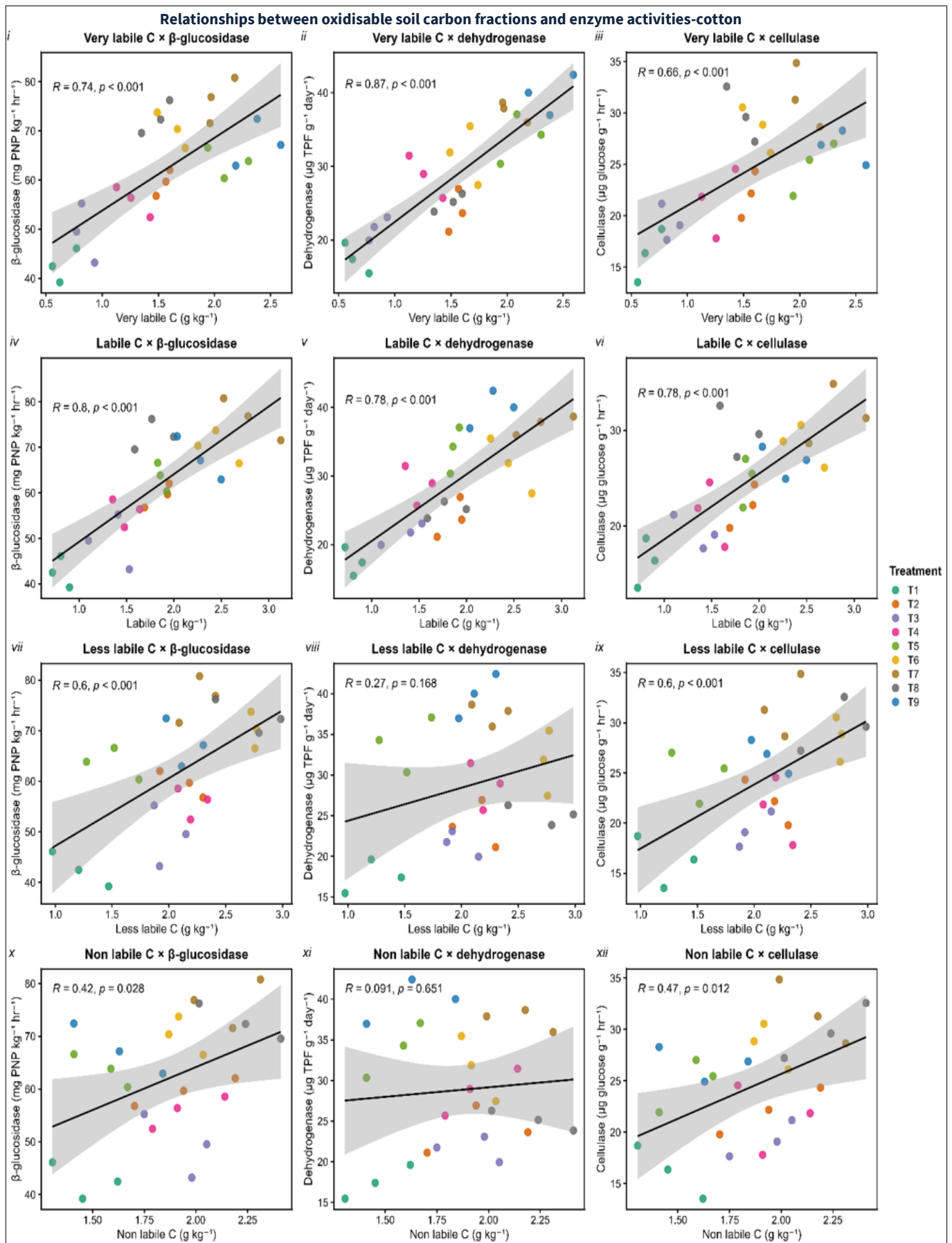


Fig. 8. Scatter plots illustrating relationships between oxidisable soil carbon fractions (very labile C, labile C, less labile C and non-labile C) and enzyme activities (β -glucosidase, dehydrogenase and cellulase) during the cotton phase. Each point represents an individual replicate ($n = 27$; 9 treatments $\times$ 3 replications), colour-coded by treatment (T1–T9), with Pearson correlation coefficients (R) and p -values indicated for each panel. Solid lines represent linear regression fits; shaded areas indicate 95 % confidence intervals. Treatments are described in Table 1.

Table 4. Goodness-of-fit statistics for full structural equation model (SEM)

Fit index	Value	Threshold	Interpretation
Chi-square (χ^2)	2.982	-	-
Degrees of freedom (df)	3	-	-
<i>p</i> -value	0.394	> 0.05	Excellent
CFI	1.000	> 0.95	Excellent
TLI	1.000	> 0.95	Excellent
RMSEA (90 % CI)	0.002 (0.000 – 0.229)	< 0.05 (Upper < 0.10)	Excellent*
SRMR	0.011	< 0.08	Acceptable

CFI: comparative fit index; TLI: Tucker-Lewis index; RMSEA: root mean square error of approximation (90 % CI); SRMR: standardised root mean square residual. Fit indices were obtained using lavaan with 2000 bootstrap iterations (ML estimator). *The wide RMSEA upper CI reflects low degrees of freedom (df = 3).

(df = 3) rather than poor model fit. By moving beyond simple correlations, the SEM allowed mechanistic hypotheses about carbon flow through the soil system to be tested and the results indicate that SOC acted mainly as a distal regulator of biological activity, influencing enzyme dynamics indirectly through biologically active intermediates. All path coefficients, specific indirect effects and variance explained values are reported in Table 5 and the simple mediation model is presented in Supplementary Table 5.

The SOC exerted significant positive direct effects on both MBC ($\beta^* = 0.595$, $p < 0.001$) and POXC ($\beta^* = 0.596$, $p < 0.001$), explaining 36.6 % and 85.8 % of variance, respectively. The MBC also showed a significant positive direct effect on POXC ($\beta^* = 0.437$, $p < 0.001$), consistent with a serial pathway in which SOC was associated with POXC both directly and indirectly through MBC. These relationships support previous findings that labile carbon fractions are more closely associated with microbial processes than total SOC (10, 11) and agree with SEM-based evidence identifying microbial biomass as a major intermediary linking SOC with soil biological functioning (41).

Table 5. Full structural equation model: Direct paths, specific indirect effects and variance explained (R^2) for soil carbon and enzyme activity relationships (n = 54)

Path (label)	β^*	SE	z	p	95 % CI [lower, upper]	Sig.
Direct paths						
SOC → MBC (a1)	0.595	0.094	6.305	< 0.001	[0.403, 0.772]	***
SOC → POXC (a2)	0.596	0.066	9.020	< 0.001	[0.478, 0.741]	***
MBC → POXC (a3)	0.437	0.063	6.958	< 0.001	[0.312, 0.560]	***
MBC → DHA (b1)	0.737	0.086	8.581	< 0.001	[0.577, 0.922]	***
POXC → DHA (b2)	0.234	0.084	2.794	0.005	[0.067, 0.399]	**
POXC → BGA (b4)	0.521	0.188	2.772	0.006	[0.176, 0.895]	**
SOC → BGA (b5)	0.316	0.169	1.871	0.061	[0.041, 0.639]	ns
POXC → CA (b7)	0.546	0.174	3.134	0.002	[0.223, 0.914]	***
SOC → CA (b8)	0.270	0.179	1.514	0.130	[-0.073, 0.622]	ns
Specific indirect effects (bootstrap 95 % CI)						
SOC → MBC → DHA (a1 × b1)	0.439	0.072	6.085	< 0.001	[0.3109, 0.5905]	***
SOC → POXC → DHA (a2 × b2)	0.139	0.052	2.680	0.007	[0.0377, 0.2418]	**
SOC → POXC → BGA (a2 × b4)	0.310	0.122	2.535	0.011	[0.1001, 0.5754]	*
SOC → POXC → CA (a2 × b7)	0.325	0.114	2.856	0.004	[0.1370, 0.5928]	**
MBC → POXC → DHA (a3 × b2)	0.102	0.040	2.517	0.012	[0.0343, 0.1942]	*
MBC → POXC → BGA (a3 × b4)	0.228	0.080	2.817	0.005	[0.0809, 0.3965]	**
Total indirect: SOC → DHA (a1b1 + a2b2 + a1a3b2)	0.639	0.086	7.385	< 0.001	[0.4514, 0.7944]	***
Variance explained (R^2)						
MBC	0.366	36.6 %			SOC (direct path a1)	
POXC	0.858	85.8 %			SOC (a2) + MBC (a3)	
DHA	0.871	87.1 %			MBC (b1) + POXC (b2)	
BGA	0.655	65.5 %			POXC (b4) + SOC (b5) direct ns ($p = 0.061$)	
CA	0.627	62.7 %			POXC (b7); SOC→CA direct ns ($p=0.130$)	

SOC: soil organic carbon; MBC: microbial biomass carbon; POXC: permanganate oxidizable carbon; DHA: dehydrogenase activity; BGA: β -glucosidase activity; CA: cellulase activity. β^* : standardised path coefficient (lavaan); SE: standard error; z: z-statistic; 95 % CI: bootstrap confidence interval; R^2 : variance explained. Path labels (a1, a2, a3, b1, b2, b4, b5, b7, b8) denote model relationships. Significance: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; ns = not significant.

For dehydrogenase, MBC was the dominant direct driver ($\beta^* = 0.737$, $p < 0.001$), with POXC contributing an additional effect ($\beta^* = 0.234$, $p = 0.005$). The SOC → MBC → DHA pathway ($\beta^* = 0.439$) was substantially stronger than the SOC → POXC → DHA pathway ($\beta^* = 0.139$). Together, these pathways explained 87.1 % of variance in DHA. The dominance of the MBC-mediated route underscores the primary role of microbial biomass in regulating oxidative enzyme activity, consistent with evidence that microbial biomass drives enzyme-mediated biochemical transformations and that both microbial growth and labile carbon availability are needed to enhance biological fertility (47).

The POXC was the primary direct driver of both BGA ($\beta^* = 0.521$, $p = 0.006$) and CA ($\beta^* = 0.546$, $p = 0.002$), explaining 65.5 and 62.7 % of their respective variances. In contrast, the direct associations of SOC with BGA ($\beta^* = 0.316$, $p = 0.061$) and CA ($\beta^* = 0.270$, $p = 0.130$) were non-significant, indicating that the associations between SOC and these extracellular enzymes were mediated through POXC. This agrees with previous observations that biologically accessible carbon fractions are more closely associated with extracellular enzyme activity than total SOC (48). This is consistent with empirical findings that POXC-type labile fractions associate more closely with enzyme activity than total SOC (10, 11).

The simple mediation model (Supplementary Table 5) indicated strong evidence of complete mediation of the SOC effect on DHA through MBC. The direct path from SOC to DHA was non-significant ($\beta^* = 0.082$, $p = 0.194$), whereas the indirect effect via MBC was significant ($\beta^* = 0.520$, $p < 0.001$), accounting for 86.5 % of the total SOC effect on DHA (total effect $\beta^* = 0.601$, $p < 0.001$). In the full SEM (Table 5), three specific indirect pathways from SOC to DHA were statistically significant. SOC → MBC → DHA ($\beta^* = 0.439^{***}$), SOC → POXC → DHA ($\beta^* = 0.139^{**}$) and MBC → POXC → DHA ($\beta^* = 0.102^*$).

The total indirect effect of SOC on DHA was $\beta^* = 0.639$ ($p < 0.001$). For BGA and CA, SOC exerted significant indirect effects via POXC. The SEM illustrating these pathways is presented in Fig. 9. Residual covariance between BGA and CA suggests coordinated polysaccharide degradation driven by substrate availability, consistent with the shared functional roles of these enzymes and with earlier findings, states that extracellular enzyme production is more closely associated with biologically available carbon than with total SOC (49). Collectively, these findings indicate that improving biological fertility in dryland Vertisols depends not only on increasing SOC but also on enhancing active carbon pools and the living microbial biomass that mediates carbon transformation.

Treatment grouping based on soil carbon and biological activity profiles

Hierarchical clustering of the nine treatments based on all measured soil carbon and biological activity parameters resolved three distinct clusters (Fig. 10). Cluster 1, comprising the control (T1), 100 % RDF (T2), 50 % RDF (T3) and 50 % N_{CR} (T4), was characterised by comparatively lower SOC, MBC, POXC, VLC and enzyme activities. Although 100 % RDF maintained moderate SOC through crop biomass returns, limited labile carbon inputs restricted carbon quality, while the inclusion of 50 % N_{CR} reflected the slower decomposition and poor nutrient synchronisation of CR under semi-arid Vertisols (24). Cluster 2, comprising 50 % N_{FM} (T5)

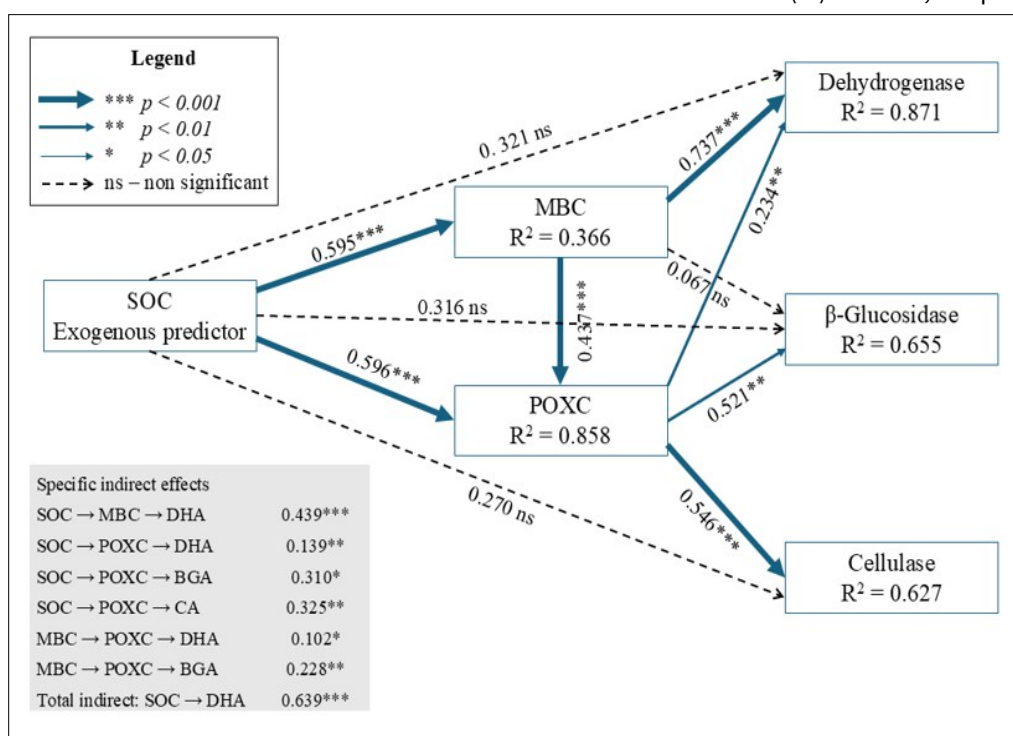


Fig. 9. Structural equation model (SEM) depicting the direct and indirect effects of soil organic carbon (SOC) on enzyme activities (dehydrogenase, β -glucosidase and cellulase) mediated through microbial biomass carbon (MBC) and permanganate oxidisable carbon (POXC). Standardised path coefficients are shown alongside arrows; arrow thickness and style indicate significance levels (solid: significant; dashed: non-significant). Coefficient of determination (R^2) values denote the proportion of variance explained for each endogenous variable. Specific and total indirect effects are listed in the inset table.

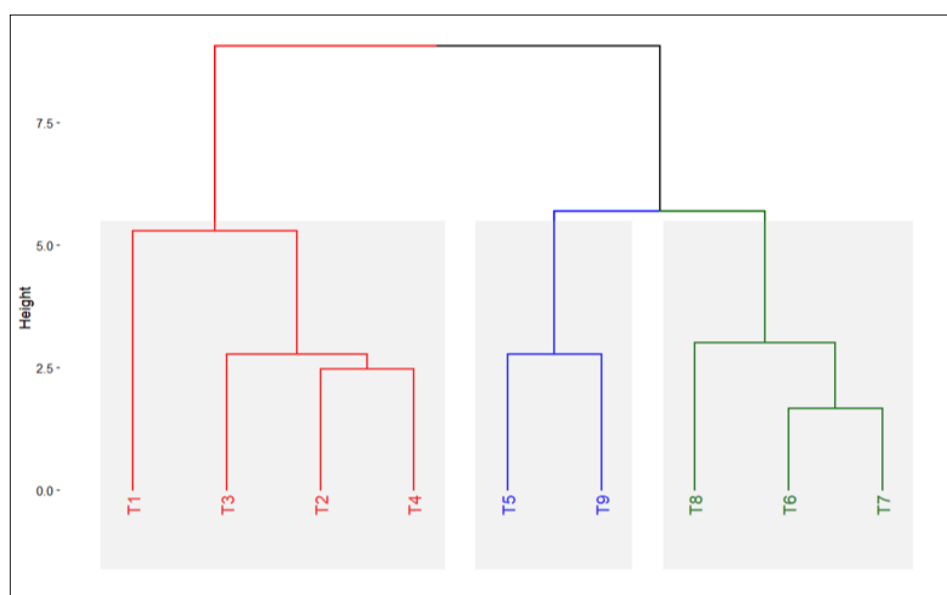


Fig. 10. Dendrogram from hierarchical clustering analysis of nine treatments (T1–T9) based on soil carbon and biological activity parameters. Three distinct clusters were identified: Cluster 1 (red; T1, T2, T3 and T4); Cluster 2 (blue; T5 and T9) and Cluster 3 (green; T6, T7 and T8), reflecting similarities in overall soil carbon and enzyme profiles among treatments. Treatments are described in Table 1.

and FYM (T9), was characterised by high VLC, MBC and DHA, reflecting enrichment of biologically active carbon pools and rapid microbial turnover, consistent with observations in organically managed tropical soils (25). Cluster 3 comprised INM_{CR} (T6), INM_{FYM} (T7) and RDF + Zn (T8), which together achieved high SOC, high LC and strong enzyme activities and thus the most balanced carbon and biological functioning of the three groups. This balance is driven by integrated nutrient inputs and nutrient-mediated stimulation of microbial activity, corroborating previous reports, found that combined FYM and fertiliser inputs raise SOC and enzyme activity relative to sole inorganic fertilisation (43). Although RDF + Zn supplies no direct organic carbon, its grouping with the INM treatments indicates that balanced zinc-supplemented fertilisation can lift soil carbon dynamics and enzyme activity to comparable levels, which is consistent with zinc acting as an enzyme cofactor that stimulates microbial activity (50) and with zinc modifying rhizosphere microbial processes that influence below-ground carbon dynamics (39).

The three clusters distinguished contrasting carbon management pathways. Cluster 2 was associated with rapid carbon turnover and microbial activity, whereas cluster 3 combined carbon accumulation, stabilisation and biological activity. In contrast, cluster 1 reflected limited carbon quality and biological activity. The clustering further indicates that FYM preferentially enriched active carbon pools, whereas integrated nutrient management provided the most balanced improvement in carbon quantity, carbon quality and biological fertility.

Conclusion

This study provides a comprehensive assessment of the long-term effects of diverse nutrient management practices on soil carbon fractions, enzyme activity and carbon management quality in a rainfed Vertisol of a semi-arid region. Four decades of contrasting management sharply separated the organic and inorganic carbon pools. Integrated nutrient management with farmyard manure (INM_{FYM}) and sole FYM built the highest SOC, whereas inorganic-dominant treatments (Recommended dose of fertilisers (RDF) + zinc (Zn) and 100 % RDF) favoured soil inorganic carbon, confirming that the two pools are governed by fundamentally different drivers. The FYM preferentially enriched the very labile fractions, while RDF + Zn, INM_{CR} and INM_{FYM} enhanced the more stable (less labile and non-labile) carbon. The CMI peaked under FYM and INM_{FYM} and its consistent decline from maize to cotton reinforced the need for assessment across successive cropping phases of carbon quality. The structural equation modelling (SEM) revealed that SOC influences enzyme activities predominantly through active carbon pools. Microbial biomass carbon (MBC) mediated the soil organic carbon (SOC) effect on dehydrogenase activity (DHA), while permanganate oxidisable carbon (POXC) was the primary driver of β -glucosidase activity (BGA) and cellulase activity (CA), supporting the substrate induction hypothesis that extracellular enzyme production is governed by biologically accessible carbon substrates rather than bulk SOC stocks. Hierarchical clustering consolidated these findings into three distinct typologies, with INM and RDF + Zn treatments forming a high-quality cluster, indicating that a steady supply of active carbon delivered directly through organic amendments or indirectly via fertiliser-driven biomass are indispensable for maintaining biological soil fertility in dryland

Vertisols. This study demonstrates that the long-term fertility of semi-arid Vertisols depends not only on the magnitude of the soil carbon stock, but also on the carbon quality and microbial functionality. The findings from this four-decade-long field experiment provide strong evidence that FYM and INM are effective strategies for maintaining active soil carbon pools, enhancing microbial activity and sustaining soil fertility. Incorporating these practices into soil health and carbon sequestration policies and leveraging their reliance on locally available resources to reduce fertiliser input costs can therefore promote long-term productivity, strengthen climate resilience and support the sustainable management of rainfed Vertisols.

Acknowledgements

The authors gratefully acknowledge the Department of Soil Science and Agricultural Chemistry, Tamil Nadu Agricultural University (TNAU), Coimbatore, for providing the necessary laboratory facilities and academic support to carry out this research. They also sincerely thank the Agricultural Research Station (ARS), Kovilpatti, for facilitating both field and laboratory facilities and extending technical assistance throughout the study. The authors further acknowledge the financial support provided under the All India Coordinated Research Project for Dryland Agriculture (AICRPDA)-National Innovations on Climate Resilient Agriculture (NICRA) scheme for sustaining the permanent manurial experiment (PME).

Authors' contributions

SKS conceived and designed the study, conducted the investigation, performed the formal analysis and prepared the original draft of the manuscript. SBK contributed to methodology and supervision. BBS contributed to conceptualisation, supervision and resource provision. SCS assisted with data curation. BK contributed to resources. GG assisted with field experimentation and data collection. SV, MM, JVNSP, VKS and PPG contributed to resources, provided technical support and reviewed and edited the manuscript. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflicts of interest to declare.

Ethical issues: None

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve the language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

1. Lehmann J, Hansel CM, Kaiser C, Kleber M, Maher K, Manzoni S, et al. Persistence of soil organic carbon caused by functional complexity. *Nat Geosci.* 2020;13(8):529–34. <https://doi.org/10.1038/s41561-020-0612-3>
2. Lal R. Soil organic matter and water retention. *Agron J.* 2020;112(5):3265–77. <https://doi.org/10.1002/agj2.20282>

3. Masto RE, Chhonkar PK, Singh D, Patra AK. Alternative soil quality indices for evaluating the effect of intensive cropping, fertilisation and manuring for 31 years in the semi-arid soils of India. *Environ Monit Assess.* 2008;136(1):419–35. <https://doi.org/10.1007/s10661-007-9697-z>
4. Sukumaran S, Purakayastha T, Sarkar B, Chakrabarti B, Bandyopadhyay K, Kumar D, et al. Assessment of carbon carrying capacities of Alfisols and Vertisols under long-term manuring and fertilization. *Soil Tillage Res.* 2024;238:105994. <https://doi.org/10.1016/j.still.2023.105994>
5. Raza S, Irshad A, Margenot A, Zamanian K, Li N, Ullah S, et al. Inorganic carbon is overlooked in global soil carbon research: a bibliometric analysis. *Geoderma.* 2024;443:116831. <https://doi.org/10.1016/j.geoderma.2024.116831>
6. Chan K, Bowman A, Oates A. Oxidizable organic carbon fractions and soil quality changes in an oxic paleustalf under different pasture leys. *Soil Sci.* 2001;166(1):61–7. <https://doi.org/10.1097/00010694-200101000-00009>
7. Culman SW, Snapp SS, Freeman MA, Schipanski ME, Beniston J, Lal R, et al. Permanganate oxidizable carbon reflects a processed soil fraction that is sensitive to management. *Soil Sci Soc Am J.* 2012;76(2):494–504. <https://doi.org/10.2136/sssaj2011.0286>
8. Marzouk SH, Kwaslema DR, Omar MM, Mohamed SH. Harnessing the power of soil microbes: their dual impact in integrated nutrient management and mediating climate stress for sustainable rice crop production: a systematic review. *Heliyon.* 2025;11(1):e41158. <https://doi.org/10.1016/j.heliyon.2024.e41158>
9. Blair GJ, Lefroy RD, Lisle L. Soil carbon fractions based on their degree of oxidation and the development of a carbon management index for agricultural systems. *Aust J Agric Res.* 1995;46(7):1459–66. <https://doi.org/10.1071/AR9951459>
10. Xu CY, Du C, Jian JS, Hou L, Wang ZK, Wang Q, et al. The interplay of labile organic carbon, enzyme activities and microbial communities of two forest soils across seasons. *Sci Rep.* 2021;11(1):5002. <https://doi.org/10.1038/s41598-021-84217-6>
11. Zhang L, Chen X, Xu Y, Jin M, Ye X, Gao H, et al. Soil labile organic carbon fractions and soil enzyme activities after 10 years of continuous fertilization and wheat residue incorporation. *Sci Rep.* 2020;10(1):11318. <https://doi.org/10.1038/s41598-020-68163-3>
12. Daunoras J, Kačergius A, Gudiukaitė R. Role of soil microbiota enzymes in soil health and activity changes depending on climate change and the type of soil ecosystem. *Biology (Basel).* 2024;13(2):85. <https://doi.org/10.3390/biology13020085>
13. Walkley A, Black IA. An examination of the Degtjareff method for determining soil organic matter and a proposed modification of the chromic acid titration method. *Soil Sci.* 1934;37(1):29–38. <https://doi.org/10.1097/00010694-193401000-00003>
14. Piper C. Soil and plant analysis. *Soil Sci.* 1943;56(1):68. <https://doi.org/10.1097/00010694-194307000-00013>
15. Vance ED, Brookes PC, Jenkinson DS. An extraction method for measuring soil microbial biomass C. *Soil Biol Biochem.* 1987;19(6):703–7. [https://doi.org/10.1016/0038-0717\(87\)90052-6](https://doi.org/10.1016/0038-0717(87)90052-6)
16. Anderson TH, Domsch KH. Ratios of microbial biomass carbon to total organic carbon in arable soils. *Soil Biol Biochem.* 1989;21(4):471–9. [https://doi.org/10.1016/0038-0717\(89\)90117-X](https://doi.org/10.1016/0038-0717(89)90117-X)
17. Weil RR, Islam KR, Stine MA, Gruver JB, Samson-Liebig SE. Estimating active carbon for soil quality assessment: a simplified method for laboratory and field use. *Am J Altern Agric.* 2003;18(1):3–17. <https://doi.org/10.1079/AJAA200228>
18. Chichester F, Chaison R. Analysis of carbon in calcareous soils using a two-temperature dry combustion infrared instrumental procedure. *Soil Sci.* 1992;153(3):237–41. <https://doi.org/10.1097/00010694-199203000-00007>
19. Eivazi F, Tabatabai M. Glucosidases and galactosidases in soils. *Soil Biol Biochem.* 1988;20(5):601–6. [https://doi.org/10.1016/0038-0717\(88\)90141-1](https://doi.org/10.1016/0038-0717(88)90141-1)
20. Casida LE Jr, Klein DA, Santoro T. Soil dehydrogenase activity. *Soil Sci.* 1964;98(6):371–6. <https://doi.org/10.1097/00010694-196412000-00004>
21. Schinner F, Von Mersi W. Xylanase-, CM-cellulase- and invertase activity in soil: an improved method. *Soil Biol Biochem.* 1990;22(4):511–5. [https://doi.org/10.1016/0038-0717\(90\)90187-5](https://doi.org/10.1016/0038-0717(90)90187-5)
22. Gomez KA, Gomez AA. Statistical procedures for agricultural research. 2nd ed. New York: John Wiley & Sons; 1984.
23. Rosseel Y. lavaan: an R package for structural equation modeling. *J Stat Softw.* 2012;48(2):1–36. <https://doi.org/10.18637/jss.v048.i02>
24. Ramteke P, Gabhane V, Kadu P, Kharche V, Jadhao S, Turkhede A, et al. Long-term nutrient management effects on organic carbon fractions and carbon sequestration in Typic Haplusterts soils of Central India. *Soil Use Manag.* 2024;40(1):e12950. <https://doi.org/10.1111/sum.12950>
25. Rai AK, Basak N, Dixit AK, Rai SK, Das SK, Singh J, et al. Changes in soil microbial biomass and organic C pools improve the sustainability of perennial grass and legume system under organic nutrient management. *Front Microbiol.* 2023;14:1173986. <https://doi.org/10.3389/fmicb.2023.1173986>
26. Liu W, Yang Z, Ye Q, Peng Z, Zhu S, Chen H, et al. Positive effects of organic amendments on soil microbes and their functionality in agro-ecosystems. *Plants (Basel).* 2023;12(22):3790. <https://doi.org/10.3390/plants12223790>
27. Preeti, Sheoran S, Prakash D, Ankit, Todarmal, Rani S, et al. Soil carbon, micronutrients and microbiological dynamics under cash crop-based cropping systems in semi-arid National Capital Region of India. *Sci Rep.* 2026. <https://doi.org/10.1038/s41598-026-35142-z>
28. Pilli K, Dash B, Sahu B, M J, Sridhar D. Soil inorganic carbon in dry lands: an unsung player in climate change mitigation. In: *Enhancing resilience of dryland agriculture under changing climate: interdisciplinary and convergence approaches.* Singapore: Springer; 2023. p. 247–57. https://doi.org/10.1007/978-981-19-9159-2_14
29. Zhou J, Shao G, Li L, Yang X, Zamanian K, Alharbi SA, et al. The over-estimation of long-term mineral fertilizer on CO₂ release from soil carbonates. *Agric Ecosyst Environ.* 2025;392:109737. <https://doi.org/10.1016/j.agee.2025.109737>
30. Kim AR, Lee CR, Ku HH. Effects of soil chemical properties on inorganic and organic carbon contents in organic, calcareous and reclaimed-tidal soils. *Korean J Soil Sci Fert.* 2025;58(1):14–26. <https://doi.org/10.7745/KJSSF.2025.58.1.014>
31. Naorem A, Jayaraman S, Dalal RC, Patra A, Rao CS, Lal R. Soil inorganic carbon as a potential sink in carbon storage in dryland soils: a review. *Agriculture (Basel).* 2022;12(8):1256. <https://doi.org/10.3390/agriculture12081256>
32. Gabhane V, Ramteke P, Chary GR, Patode R, Ganvir M, Chorey A, et al. Effects of long-term nutrient management in semi-arid Vertisols on soil quality and crop productivity in a cotton-greengram intercropping system. *Field Crops Res.* 2023;303:109115. <https://doi.org/10.1016/j.fcr.2023.109115>
33. Zhang Z, Yan J, Han X, Zou W, Chen X, Lu X, et al. Labile organic carbon fractions drive soil microbial communities after long-term fertilization. *Glob Ecol Conserv.* 2021;32:e01867. <https://doi.org/10.1016/j.gecco.2021.e01867>
34. Zhou Z, Zhang S, Jiang N, Xiu W, Zhao J, Yang D. Effects of organic fertilizer incorporation practices on crops yield, soil quality and soil fauna feeding activity in the wheat-maize rotation system. *Front Environ Sci.* 2022;10:1058071. <https://doi.org/10.3389/fenvs.2022.1058071>
35. Datta S, Mazumdar SP, Majumdar B, Alam N, Chattopadhyay L, Ghosh S, et al. Impact of integrated nutrient management on soil microbiome diversity and health in rice based cropping system: insights from long-term agricultural practices. *Rhizosphere.* 2025;33:101048. <https://doi.org/10.1016/j.rhisph.2025.101048>
36. Woodings FS, Margenot AJ. Revisiting the permanganate oxidizable carbon (POXC) assay assumptions: POXC is lignin sensitive. *Agric Environ Lett.* 2023;8(1):e20108. <https://doi.org/10.1002/acl.20108>
37. Li T, Zhang Y, Bei S, Li X, Reinsch S, Zhang H, et al. Contrasting impacts of manure and inorganic fertilizer applications for nine years on soil organic carbon and its labile fractions in bulk soil and

- soil aggregates. *Catena*. 2020;194:104739. <https://doi.org/10.1016/j.catena.2020.104739>
38. Aher SB, Lakaria BL. Impact of organic and inorganic nutrient management on soil carbon fractions and indices under soybean-wheat cropping in a Vertisol. *Soil Use Manag*. 2025;41(4):e70149. <https://doi.org/10.1111/sum.70149>
 39. Lv H, Ji C, Ding J, Yu L, Cai H. High levels of zinc affect nitrogen and phosphorus transformation in rice rhizosphere soil by modifying microbial communities. *Plants (Basel)*. 2022;11(17):2271. <https://doi.org/10.3390/plants11172271>
 40. Zhang M, Han J, Jiao J, Han J, Zhao X, Hu K, et al. Soil carbon management index under different land use systems and soil types of Sanjiang Plain in Northeast China. *Agronomy (Basel)*. 2023;13(10):2533. <https://doi.org/10.3390/agronomy13102533>
 41. Ghosh T, Sharma S, Walia S. Soil enzymes stoichiometry and microbial metabolic limitation under diverse nutrient management practices in rice-wheat cropping system: insight from a 40-year long-term field experiment. *Total Environ Microbiol*. 2025:100030. <https://doi.org/10.1016/j.temicr.2025.100030>
 42. Li Y, Liu X, Lu J, Feng H, Chen J, Yang Q, et al. Soil extracellular enzymes, soil carbon and nitrogen storage under straw return: a data synthesis. *Ind Crops Prod*. 2025;228:120884. <https://doi.org/10.1016/j.indcrop.2025.120884>
 43. Kumari M, Sheoran S, Prakash D, Yadav DB, Yadav PK, Jat MK. Long-term application of organic manures and chemical fertilizers improve the organic carbon and microbiological properties of soil under pearl millet-wheat cropping system in North-Western India. *Heliyon*. 2024;10(3):e25333. <https://doi.org/10.1016/j.heliyon.2024.e25333>
 44. Genova G, Borruso L, Signorini M, Mitterer M, Niedrist G, Cesco S, et al. Analyzing soil enzymes to assess soil quality parameters in long-term copper accumulation through a machine learning approach. *Appl Soil Ecol*. 2024;195:105261. <https://doi.org/10.1016/j.apsoil.2023.105261>
 45. Qu Q, Wang Z, Gan Q, Liu R, Xu H. Impact of drought on soil microbial biomass and extracellular enzyme activity. *Front Plant Sci*. 2023;14:1221288. <https://doi.org/10.3389/fpls.2023.1221288>
 46. Sinsabaugh RL, Follstad Shah JJ. Ecoenzymatic stoichiometry and ecological theory. *Annu Rev Ecol Evol Syst*. 2012;43(1):313–43. <https://doi.org/10.1146/annurev-ecolsys-071112-124414>
 47. Jing Y, Wenhao Y, Fan W, Biqing Z, Shihe X. Driving factors of soluble organic nitrogen dynamics in paddy soils: structure equation modeling analysis. *Pedosphere*. 2020;30(6):801–9. [https://doi.org/10.1016/S1002-0160\(18\)60032-3](https://doi.org/10.1016/S1002-0160(18)60032-3)
 48. Jin Z, Ping L, Wong M. Systematic relationship between soil properties and organic carbon mineralization based on structural equation modeling analysis. *J Clean Prod*. 2020;277:123338. <https://doi.org/10.1016/j.jclepro.2020.123338>
 49. Tang J, Zhao J, Qin Z, Chen L, Song X, Ke Q, et al. Structural equation model was used to evaluate the effects of soil chemical environment, fertility and enzyme activity on eucalyptus biomass. *R Soc Open Sci*. 2023;10(4):221570. <https://doi.org/10.1098/rsos.221570>
 50. Liu YM, Cao WQ, Chen XX, Yu BG, Lang M, Chen XP, et al. The responses of soil enzyme activities, microbial biomass and microbial community structure to nine years of varied zinc application rates. *Sci Total Environ*. 2020;737:140245. <https://doi.org/10.1016/j.scitotenv.2020.140245>

Additional information

Peer review: Publisher thanks Sectional Editor and the other anonymous reviewers for their contribution to the peer review of this work.

Reprints & permissions information is available at https://horizonpublishing.com/journals/index.php/PST/open_access_policy

Publisher's Note: Horizon e-Publishing Group remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Indexing: Plant Science Today, published by Horizon e-Publishing Group, is covered by Scopus, Web of Science, BIOSIS Previews, Clarivate Analytics, NAAS, UGC Care, etc
See https://horizonpublishing.com/journals/index.php/PST/indexing_abstracting

Copyright: © The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution and reproduction in any medium, provided the original author and source are credited (<https://creativecommons.org/licenses/by/4.0/>)

Publisher information: Plant Science Today is published by HORIZON e-Publishing Group with support from Empirion Publishers Private Limited, Thiruvananthapuram, India.